

Illiquidity and the cost of equity capital: Evidence from actual estimates of capital cost for U.S. data

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Abstract

Illiquidity measures appear to be related to monthly realized returns but do they impact long-run costs of capital (CoC) for firms? Using U.S. data, we find cross-sectional evidence that, controlling for market capitalization, the Amihud (2002, *Journal of Financial Markets*, 5, 31) measure of illiquidity is negatively related to CoC estimates. A difference-in-differences analysis around exogenous brokerage closures reveals that Amihud illiquidity increases without an impact on CoC. Nonetheless, other illiquidity measures, such as those based on serial covariances, zero returns, and price impact, do show a strong positive relation with CoC. However, we do not find evidence that liquidity risk and the probability of informed trade influence CoC. Overall, our results advance our understanding of precisely which illiquidity measures influence required firm returns.

KEYWORDS

cost of capital, determinants of equity returns, liquidity Premia, trading costs

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Do investors demand higher returns from less liquid securities? This is an intriguing question in financial economics and has attracted much attention. In their seminal work, Amihud and Mendelson (1986) show that asset returns are positively related to the quoted bid-ask spread. Using an alternative measure based on price moves per unit volume, Amihud (2002) confirms that liquidity measures are indeed positively related to an average realized returns.¹ These findings suggest that the answer to the posed question is in the affirmative. Pointing to the centrality of this result in finance research, the papers of Amihud and Mendelson (1986) and Amihud (2002) have garnered more than 15,000 citations on Google Scholar® to date.

The method used for estimating the cross-sectional liquidity-return relation in the literature has generally been to regress monthly realized returns on an illiquidity measure lagged by 1 or 2 months. The argument is that illiquidity raises the required return, which manifests itself in monthly average returns. Extending this argument, Amihud and Mendelson (2008) propose that, since required returns are proxies for costs of capital, corporations should be concerned about illiquidity having a potentially deleterious effect on such costs.²

The average monthly returns used to measure the illiquidity–return relation are not directly related to the cost of capital (or required return) for the typical firm's projects.³ Most corporations are concerned more about capital costs for projects that take years to pay off. As such, the issue of whether illiquidity affects corporations' required returns should involve a measurement of the relation between illiquidity and a measure of the internal rate of return on firms'

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longer-term cash flows. Happily, such cost of capital measures is readily available in the existing literature. These measures combine analysts' forecasts and financial statements to back out a long-term discount rate on firm's equity, and this rate, in excess of a benchmark Treasury bond yield, is the implied equity cost of capital (*ICC*). The goal of our analysis is to explore the relationship between equity illiquidity and *ICC* estimates for U.S. data.

When we perform a portfolio-based analysis and consider double sorts by size (using market capitalization as the proxy for size) and the standard Amihud (2002) measure of illiquidity, we find that the cost of capital is *lower* for portfolios with higher levels of illiquidity in all four size-based portfolios. For each size group, stocks in the highest Amihud-based group have a cost of capital that is at least 0.7% lower than that for stocks in the group with the lowest levels of the Amihud measure. Thus, we get a result opposite of what we expect a priori. We also conduct Fama and MacBeth (1973) (FM)-type regressions using *ICC* estimates as the dependent variable. In this analysis, we again find that Amihud illiquidity is negatively related to firms' cost of equity capital, controlling for previously known determinants of *ICC*. Regressions where variables are reduced to a rank ordering yield qualitatively similar results: that Amihud illiquidity is negatively related to *ICC*.

We are cognizant that *ICC* estimation is reliant on analysts' forecasts but given that the literature extensively uses these as inputs to the cost of capital estimates (Easton, 2009), we feel justified in simply extending this literature to illiquidity pricing. But we do perform robustness checks with alternative *ICC* measures. Specifically, while for the most part, we use *ICC* adapted from Pástor et al. (2008), we report similar conclusions for alternative analysts-based *ICC* computations developed by Easton (2004) and Ohlson and Juettner-Nauroth (2005).

Market capitalization (size) plays a crucial role in our cross-sectional regressions. This control is desirable because Lee et al. (2009) have already shown that market capitalization is negatively related to *ICC*. Further, since both market capitalization and the Amihud illiquidity measure are highly skewed, it is desirable to use functional forms that reduce the influence of extreme size and illiquidity observations. We, therefore, follow Lou and Shu (2017) and Amihud and Noh (2021) and use the logarithms of size and Amihud in our regressions. While these variables are highly negatively correlated in the cross-section, we show that it is only the Amihud illiquidity that flips sign from positive to negative when it is included along with market capitalization, suggesting that it noisily proxies for (inverse) size when size is excluded.

We consider alternative illiquidity measures; specifically, the original measure, the quoted spread, considered by Amihud and Mendelson (1986), the Fong et al. (2017) measure of proportion of zero days, a version of the Brennan and Subrahmanyam (1996) price impact measure, λ , and a modified Roll (1984) measure computed by Abdi and Ranaldo (2017). Interestingly, we find reliable evidence that all these measures are strongly positively priced in *ICC*. The evidence of their pricing in *ICC* is *stronger* than that in realized monthly returns.

We further extend our analysis to account for liquidity risk. Specifically, we consider whether the standard measures of liquidity risk developed by Acharya and Pedersen (2005) as well as Pástor and Stambaugh (2003) are related to *ICC*. Finally, we consider whether higher values of the information risk measure *PIN* (Easley et al., 2002, as modified by Duarte et al., 2020) lead to higher values of *ICC*. We find no consistent evidence that liquidity risk, or *PIN* is reliably related to the cost of capital estimates.

To effectively address the joint determination of *ICC* and liquidity, we next perform a difference-in-differences (DiD) estimation around the exogenous brokerage closures described in Kelly and Ljungqvist (2012).⁴ These authors show that the closures led to a loss of sell-side analyst coverage and made the affected stocks less liquid (based on the Amihud measure). In our sample, we first verify the basic Kelly and Ljungqvist (2012) finding of decreased liquidity following the closures. We then examine how *ICC* behaves in the same event windows for treated and control stocks. We find no evidence of an increase in *ICC* for the treated stocks relative to the controls; in fact, the point estimates show a decrease instead, which is not statistically significant. Thus, the DiD event does not confirm the conjecture that a change in Amihud illiquidity influences corporations' cost of capital.

We are aware of the paper by Saad and Samet (2017), who show that cost of capital estimates are positively related to illiquidity, using panel data across several countries. We instead use U.S. data to benchmark our work against the extensive work done on domestic illiquidity premia. Nonetheless, we point out another key difference: Saad and Samet (2017) use the book value of assets as a measure of firm size, whereas we use market capitalization, to be consistent with Amihud and Noh (2021). We confirm Lee et al. (2009) finding that market capitalization is inversely related to *ICC*⁵ and do not find evidence that the Amihud measure is related to *ICC*, though there is evidence of illiquidity pricing for other measures. Overall, our study sheds useful light on which illiquidity measures influence firms' cost of capital.

1 | IMPLIED COST OF CAPITAL

In this section, we describe the methodology used to estimate the firm-level implied cost of capital (*ICC*). Our estimation of firm-level *ICC* follows the approach of Li et al. (2013).⁶ The firm-level *ICC* is constructed as the internal rate of return that equates the present value of future dividends/free cash flows to the current stock price⁷:

$$P_t = \sum_{k=1}^{\infty} \frac{E_t(FCFE_{t+k})}{(1+r_e)^k}, \quad (1)$$

where P_t is the current stock price, $E_t(FCFE_{t+k})$ is the expected future free cash flows to equity for period $t+k$ conditional on information available at time t , and r_e is the long-term cost of equity capital at time t .

There are two key assumptions in our empirical implementation of the free cash flow model: (a) short-run earnings growth rates converge in the long run to the growth rate of the overall economy and (b) competition drives economic profits on new investments to zero in the long-run (the marginal rate of return on investment—the ROI on the next dollar of investment—will converge to the cost of capital). We use these assumptions to forecast earnings growth rates and free cash flows during the transition period from the short-run to the long-run steady-state.

We implement Equation (1) in two parts: (i) the present value of free cash flows up to a terminal period $t+T$ and (ii) a terminal value that captures the present value of free cash flows beyond the terminal period. We estimate free cash flows up to year $t+T$, as the product of annual earnings forecasts and one minus the plowback rate:

$$E_t(FCFE_{t+k}) = FE_{t+k} \times (1 - b_{t+k}), \quad (2)$$

where FE_{t+k} and b_{t+k} are earnings forecasts and plowback rate forecasts for year $t+k$, respectively. Beyond $t+T$, we assume that future investments produce zero economic profits and thus apply a steady-state perpetuity formula to determine terminal value. The value of T , following Li et al. (2013), is set to be 15 years. *ICC* is then determined as the discount rate that satisfies Equation (1). We present full details of our *ICC* computations in the Appendix. Our basic premise is that if illiquidity is a consideration in setting long-term discount rates for corporations, then higher illiquidity should imply higher *ICC* in the cross-section. We use our panel of monthly *ICC* estimates to formally test this hypothesis.

2 | DATA AND SUMMARY STATISTICS

For empirical tests involving *ICC*, our sample begins in January 1977 due to the availability of I/B/E/S analyst consensus earnings forecasts. Our sample period ends in December 2021. Our primary sample consists of NYSE/AMEX stocks. However, we examine the robustness of our key findings with a “holdout” sample of NASDAQ stocks in Section 5. The previous section described the methodology for computing *ICC*, and now we turn to describe the other key variables used in our empirical analysis.

Amihud is the Amihud (2002) measure of illiquidity calculated at the end of every month using daily return and dollar trading volume over the prior 12 months (with minimum of 200 trading days while excluding days with trading volume below 100 shares). It is defined as the average of the ratio of absolute return to dollar trading volume for a given stock i at the end of month t :

$$ILLIQ_{it} = \frac{1}{D_{it}} \sum_{d=1}^{D_{it}} \frac{|R_{id}|}{VOLD_{id}}, \quad (3)$$

where D_{it} is the number of eligible trading days over the prior 12 months, R_{id} is the daily stock return and $VOLD_{id}$ is the daily dollar trading volume. The ratio is a measure of the daily price impact of order flow. Initially, we focus on our analysis of Amihud illiquidity since it is the most widely used measure of illiquidity in the literature. Later we expand our analysis to a more comprehensive set of illiquidity measures (see Section 3.3).

SIZE is the market capitalization in billions of dollars computed at the end of every month.

B/M is the ratio of book value equity to the market value of equity computed at the end of every month where the book value of equity is computed as in Fama and French (1992).

Return11 is the 11-month cumulative stock return skipping the most recent month and is a measure of past price momentum. For instance, this measure as of the end of December 2018 would be the cumulative return from January 2018 to November 2018.

Return1 is the most recent month's stock return and captures monthly return reversals as in Jegadeesh (1990).

Beta is the market beta calculated using monthly data over the last 60 months with a minimum requirement of 36 months.

GP/AT is the ratio of gross profits to total assets as in Novy-Marx (2013).

AG is the percentage growth in total assets as in Cooper et al. (2008).

The sample consists of all NYSE/AMEX stocks with prices between \$5 and \$1000. In addition, we also exclude all ADRs, REITs, closed-end funds, and unit trusts (excluding CRSP share codes above 11). The accounting variables in the list above are assumed to be known only 6 months after the fiscal year-end.

Panel A of Table 1 reports summary statistics on *Amihud* and other firm characteristics described above. The key thing to note in Panel A is that both *Amihud* and *SIZE* have very high positive (standardized) skewness with values of 5.630 and 5.331 respectively (skewness of the normal distribution is zero). This potentially induces an undue influence of extreme observations in regression analysis. However, taking natural logs reduces the skewness of both *Amihud* and *SIZE* and brings it close to zero: $\ln(\text{Amihud})$ has a skewness of 0.272, and $\ln(\text{SIZE})$ has a skewness of 0.084. This can also be seen intuitively by comparing the difference between the means and medians of both variables before and after taking natural logs. The importance of taking logs will become clearer when we examine the cross-sectional correlation between *Amihud* and *SIZE* (Table 2). We also take the natural log of $(1 + \text{AG})$ to take into account the high skewness in asset growth (skewness of 3.573 in *AG* vs. skewness of 1.598 in $\ln(1 + \text{AG})$).

Panel B of Table 1 reports annual (as of June 30) summary statistics for the implied risk premium (*ICC* – 10-year Treasury yield). For convenience, in the remainder of the paper, we continue to use the same notation (i.e., *ICC*) for this implied risk premium. The number of NYSE/AMEX firms with available *ICC* estimates increases from 686 in 1977 to around 1000 towards the end of the sample period. The equal-weighted mean *ICC* exhibits considerable time variation over time.

Table 2 reports (time-series averages of) cross-sectional correlations among *ICC* and the various firm characteristics. We report Pearson correlations below the diagonal and Spearman rank correlations in *italics* above the diagonal. The Pearson correlation between *SIZE* and *Amihud* is only -0.135 which does not seem that high, but this is distorted by the high positive skewness in both variables. The rank correlation between *Amihud* and *SIZE* which eliminates the effects of outliers and accounts for skewness, on the other hand, is a very high (in absolute value) -0.947 . The Pearson correlation increases to -0.492 if we take the logarithm of *SIZE* but leave *Amihud* untouched. The correlation increases further to -0.557 if we take the logarithm of *Amihud* but leave *SIZE* untouched. But the correlation increases to -0.940 (similar to the rank correlation) when we take logarithms of both *Amihud* and *SIZE* revealing that a stock's *Amihud illiquidity is strongly associated with its market capitalization*: large firms are highly liquid and small firms are highly illiquid. Overall, the results in Table 2 show that Pearson correlations after taking the logarithm of *Amihud* are similar to rank correlations involving *Amihud* suggesting that it is desirable to use logarithms of *Amihud* (or ranks of *Amihud*) for robust empirical findings. We take this into account in our cross-sectional asset pricing tests that examine the relationship between illiquidity and cost of capital.

ICC is positively correlated with *Amihud* and $\ln(\text{Amihud})$ suggesting that highly illiquid firms have a high cost of capital, and negatively correlated with *SIZE* and $\ln(\text{SIZE})$ suggesting large firms have a low cost of capital. Given the high negative correlation between $\ln(\text{SIZE})$ and $\ln(\text{Amihud})$, however, it is unclear based on these univariate results whether *SIZE* or *Amihud* might be a proxy of the other. We examine this more carefully using conditional portfolio sorts and cross-sectional regression tests.

In other correlations, *ICC* is positively correlated with *Beta* and $\ln(\text{B/M})$ and negatively correlated with *Return11*. The positive correlation between *ICC* and *Beta* is consistent with the predictions of the simple CAPM (unlike in the case of realized returns). The positive correlation between *ICC* and $\ln(\text{B/M})$ can be consistent with both risk and mispricing interpretations. The negative correlation between *ICC* and past price momentum, however, is inconsistent with a risk interpretation because it suggests high momentum stocks, in fact, have lower expected returns not higher expected returns. As discussed earlier, in all of the above, rank correlations are similar to those based on logs.

TABLE 1 Summary statistics.

Panel A: Summary statistics for various firm characteristics											
	<i>Amihud</i>	<i>SIZE</i>	<i>Ln(Amihud)</i>	<i>Ln(SIZE)</i>	<i>Ln(B/M)</i>	<i>Return11</i>	<i>Beta</i>	<i>GP/AT</i>	<i>Ln(1 + AG)</i>	<i>Return</i>	<i>ICC</i>
Fifth percentile	0.001	0.064	−8.235	3.703	4.870	−0.359	0.273	0.035	−0.134	−0.134	0.003
Median	0.039	1.131	−4.874	6.514	6.343	0.079	1.030	0.276	0.068	0.006	0.048
Mean	0.216	5.170	−4.741	6.542	6.268	0.140	1.079	0.317	0.098	0.012	0.060
95th percentile	1.058	22.344	−0.736	9.473	7.409	0.832	2.087	0.776	0.448	0.179	0.165
StdDev	0.513	13.556	2.279	1.753	0.791	0.401	0.551	0.238	0.198	0.099	0.050
Skewness	5.630	5.331	0.272	0.084	−0.578	1.630	0.608	1.128	1.598	0.690	1.434
Panel B: Cross-sectional distribution of ICC by year											
	5%	Median	Mean	95%	StdDev	Skewness	N				
1977	0.023	0.062	0.073	0.153	0.046	1.937	686				
1978	−0.003	0.044	0.053	0.139	0.050	2.383	807				
1979	0.005	0.054	0.062	0.140	0.053	3.566	952				
1980	−0.005	0.050	0.060	0.174	0.057	2.093	1000				
1981	−0.030	0.032	0.047	0.170	0.077	6.155	1142				
1982	−0.023	0.043	0.062	0.215	0.080	2.486	1061				
1983	−0.023	0.040	0.056	0.189	0.071	2.462	1125				
1984	−0.029	0.027	0.044	0.181	0.068	1.855	1288				
1985	−0.020	0.028	0.045	0.166	0.058	1.342	1193				
1986	−0.006	0.036	0.055	0.179	0.058	1.313	1176				
1987	−0.012	0.035	0.047	0.151	0.051	1.209	1177				
1988	−0.015	0.029	0.041	0.138	0.049	1.369	1179				
1989	−0.011	0.030	0.042	0.143	0.049	1.863	1207				
1990	−0.007	0.034	0.050	0.163	0.055	1.591	1170				
1991	−0.007	0.036	0.054	0.176	0.060	1.385	1155				
1992	0.002	0.047	0.064	0.191	0.058	1.315	1188				
1993	0.007	0.046	0.059	0.160	0.048	1.442	1321				
1994	0.000	0.036	0.049	0.154	0.046	1.422	1427				
1995	0.010	0.046	0.057	0.151	0.045	1.667	1486				
1996	−0.002	0.035	0.045	0.136	0.042	1.452	1565				
1997	−0.007	0.030	0.039	0.125	0.042	1.905	1653				

TABLE 1 (Continued)

Panel B: Cross-sectional distribution of ICC by year						
	5%	Median	Mean	95%	StdDev	Skewness
N						
1998	0.003	0.040	0.052	0.146	0.044	1.665
1999	−0.003	0.040	0.050	0.139	0.045	1.423
2000	−0.007	0.049	0.059	0.158	0.053	1.766
2001	−0.002	0.040	0.052	0.157	0.048	1.516
2002	0.005	0.044	0.058	0.160	0.048	1.351
2003	0.012	0.050	0.061	0.153	0.044	1.524
2004	0.004	0.037	0.049	0.143	0.043	1.752
2005	0.015	0.046	0.057	0.145	0.041	2.097
2006	0.003	0.041	0.052	0.139	0.043	2.030
2007	0.002	0.036	0.046	0.129	0.039	1.626
2008	0.011	0.051	0.063	0.161	0.048	1.968
2009	0.007	0.059	0.074	0.196	0.059	1.426
2010	0.022	0.068	0.081	0.182	0.050	1.367
2011	0.021	0.066	0.078	0.179	0.050	1.491
2012	0.039	0.083	0.095	0.191	0.048	1.410
2013	0.023	0.062	0.072	0.157	0.044	1.737
2014	0.022	0.058	0.069	0.156	0.042	1.570
2015	0.017	0.056	0.065	0.145	0.040	1.544
2016	0.033	0.069	0.080	0.165	0.044	1.927
2017	0.023	0.056	0.068	0.164	0.044	1.890
2018	0.018	0.056	0.067	0.163	0.045	1.807
2019	0.026	0.068	0.082	0.195	0.054	1.878
2020	0.035	0.096	0.114	0.251	0.068	1.050
2021	0.026	0.063	0.081	0.180	0.115	22.723
						958

Note: ICC is computed based on the 15-year discounted cash flow method (please see the text and the Appendix for more details). We subtract the yield on 10-year treasuries to compute the implied premium. The ICC in Panels A and B refer to this implied premium. In Panel A, *Amihud* is the Amihud (2002) illiquidity calculated using 1 year of data (using a minimum of 200 days with available data) as the average of the ratio of absolute return to dollar volume. *SIZE* is the market capitalization in billions of dollars. *B/M* is the ratio of book equity to market capitalization, where book equity is calculated as in Fama and French (1993). *Return11* is the 11-month cumulative return skipping 1 month. *Beta* is the market beta calculated using monthly data over the past 60 months (36 months minimum). *GP/AT* is the ratio of gross profit to total assets. *AG* is the percentage growth in total assets. We use $Ln(1 + AG)$ to reduce the skewness in *AG*. All accounting variables are assumed to be known 6 months after the fiscal year-end. We compute cross-sectional statistics (mean, standard deviation, skewness, and percentiles) of each variable each month. Panel A reports the time-series averages of these cross-sectional statistics. Panel B reports year-by-year cross-sectional distribution of ICC as of June of each year (column N shows the number of stocks). The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000. The sample period is 1977–2021.

TABLE 2 Cross-sectional correlations among returns, ICC, Amihud illiquidity, and firm characteristics.

	<i>Amihud</i>	<i>SIZE</i>	<i>Ln(Amihud)</i>	<i>Ln(SIZE)</i>	<i>Ln(B/M)</i>	<i>Return11</i>	<i>Beta</i>	<i>GP/AT</i>	<i>Ln(1 + AG)</i>	<i>Return</i>	<i>ICC</i>
<i>Amihud</i>		−0.947	1.000	−0.947	0.248	−0.012	0.049	0.055	−0.080	0.010	0.187
<i>SIZE</i>	−0.135		−0.947	1.000	−0.326	0.092	−0.036	−0.070	0.088	0.038	−0.235
<i>Ln(Amihud)</i>	0.586	−0.557		−0.947	0.248	−0.012	0.049	0.055	−0.080	0.010	0.187
<i>Ln(SIZE)</i>	−0.492	0.627	−0.940		−0.326	0.092	−0.036	−0.070	0.088	0.038	−0.235
<i>Ln(B/M)</i>	0.132	−0.160	0.231	−0.309		−0.311	−0.086	−0.337	−0.209	−0.098	0.267
<i>Return11</i>	0.072	0.009	0.052	0.032	−0.318		−0.018	0.021	−0.034	0.002	−0.176
<i>Beta</i>	−0.052	−0.083	0.033	−0.033	−0.087	0.015		0.087	0.044	0.001	0.185
<i>GP/AT</i>	0.024	−0.010	0.043	−0.062	−0.298	0.034	0.050		0.017	0.009	−0.089
<i>Ln(1 + AG)</i>	−0.064	0.008	−0.061	0.056	−0.162	−0.043	0.058	−0.019		−0.013	−0.082
<i>Return</i>	0.053	0.001	0.046	0.007	−0.094	0.005	0.011	0.009	−0.023		−0.048
<i>ICC</i>	0.123	−0.106	0.200	−0.238	0.252	−0.140	0.167	−0.097	−0.043	−0.039	

Note: This table provides time-series averages of cross-sectional correlations among return, ICC, Amihud illiquidity, and various other firm characteristics. Variables are defined in Table 1. The upper part of the matrix (in italics) contains rank correlations. The lower part of the matrix contains correlations in levels. The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000. The sample period is 1977–2021.

3 | EMPIRICAL RESULTS

In this section, we discuss our central results connecting illiquidity to monthly realized returns and *ICC*, using portfolio and regression approaches.

3.1 | Results from size-*Amihud* illiquidity portfolios

Ever since Banz (1981) and Fama and French (1992), it has been known that market capitalization influences realized returns. Fama and French (1993) provide a risk-based interpretation for size, arguing that small firms are more likely to get distressed, and therefore command higher expected returns. As we show in the previous section, size, and *Amihud* illiquidity exhibit extremely high cross-sectional correlation. So, the question is the following: Controlling for *SIZE*, is *Amihud* positively priced in short-term returns and long-term cost of capital? We examine this issue first through portfolio tests. This analysis has the virtue of not depending on any specific functional form of illiquidity or market capitalization.

One possible approach is to form four *SIZE*-based portfolios and then divide each *SIZE* portfolio into four *Amihud* portfolios to construct a total of 16 *SIZE-Amihud* portfolios. Then, in each *SIZE* quartile, we can compute average returns and average *ICC* across the 4 *Amihud* portfolios to address our illiquidity pricing question. Given the high correlation between *SIZE* and *Amihud* ranks, however, a simple conditional bivariate sort does not suffice to control for the *SIZE* effect in *Amihud*.⁸ Sorting first on *SIZE* and then sorting again on *Amihud* potentially could sort and simply resort to *SIZE*. In a 4 × 4 bivariate conditional sort, indeed the median *SIZE* of *Amihud* portfolios (not reported in a table) declines sharply from low to high *Amihud* portfolios in every *SIZE* quartile. For instance, in the largest size quartile, the median size declines from \$53 billion to \$6 billion as we move from the lowest to the highest *Amihud* portfolio, and in the smallest size quartile between \$0.39 billion and \$0.10 billion.

We pursue a different approach to keep the median *SIZE* more stable across the four *Amihud* portfolios. First, we construct four *SIZE* portfolios, then we divide each *SIZE* portfolio into four more *SIZE* portfolios, and finally, we divide each of the second-stage *SIZE* portfolios into four *Amihud* portfolios. This generates a total of 64 *SIZE-SIZE-Amihud* portfolios. Let us denote each portfolio with the index (i, j, k) where $i = 1 - 4$ denotes the first-stage *SIZE* sort, $j = 1 - 4$, denotes the second-stage *SIZE* sort and $k = 1 - 4$ denotes the third-stage *Amihud* sort. Then for each i , we combine the same rank *Amihud* portfolios across the four second-stage *SIZE* portfolios. Thus, we combine $(i, 1, 1)$, $(i, 2, 1)$, $(i, 3, 1)$, and $(i, 4, 1)$ to construct the Low *Amihud* portfolio, $(i, 1, 2)$, $(i, 2, 2)$, $(i, 3, 2)$, and $(i, 4, 2)$ to construct *Amihud* portfolio 2, and so on. We thus collapse the 64 *SIZE-SIZE-Amihud* portfolios into 16 *SIZE-Amihud* portfolios (we go from (i, j, k) to (i, k)). This sequential sorting approach does a better job of controlling for the high correlation between *SIZE* and *Amihud* ranks and ensures that the median *SIZE* remains more stable across the four *Amihud* portfolios. Indeed, in the largest size quartile, the median size declines from \$35 billion to \$13 billion moving from the lowest to the highest *Amihud* portfolio, and in the smallest size quintile from \$0.28 billion to \$0.26 billion.

Now we turn to discuss the results. Panel A of Table 3 reports the average annualized monthly excess returns (in excess of monthly T-bill return) 2 months after the portfolio formation date for the 16 *SIZE-Amihud* portfolios. We employ this 2-month gap to minimize any microstructure concerns such as bid-ask bounce. The first row of the panel presents results based on univariate sorts of *Amihud* for comparison with the bivariate sorts. The univariate sorts show that monthly returns increase with *Amihud* but the monthly average spread of 2.37% between the High and the Low portfolio is statistically insignificant and hence not priced. The bivariate sorts show that *Amihud* is perhaps marginally priced in small stocks and not priced in any other size portfolio. In the Small portfolio, the average spread between low and high *Amihud* portfolio is 4.49% annualized and marginally significant (t statistic of 1.83).

Panel B reports results for long-term implied cost of capital. The first row provides univariate results and, not surprisingly, shows that *ICC* increases with *Amihud*. Indeed, the spread in *ICC* between High and Low portfolios is 2.58% and highly statistically significant. However, when we control for *SIZE* the results reverse. In every *SIZE* quartile, the high *Amihud* portfolio has a lower cost of capital than the low *Amihud* portfolio despite the fact that the high *Amihud* portfolios are still a bit smaller in size (smaller firms have higher *ICC* as we can see in Panel B). Specifically, in each *SIZE* quartile, the high *Amihud* portfolio has an implied cost of capital that is lower by at least 0.7% (ranging from 0.7% to 1.9%) than that of the low *Amihud* portfolio. The t statistics (computed with 36 Newey-West lags to control for the high serial dependence in *ICC*) are all statistically significant. The portfolio with the lowest average *ICC* in Panel B is the Big portfolio with High *Amihud* illiquidity.

TABLE 3 Average return and ICC for portfolios sorted by size and Amihud illiquidity.

SIZE	Ln(Amihud)				
	Low	Q2	Q3	High	High – low
Panel A: Postformation Return					
All	8.57 (4.61)	10.07 (5.47)	10.95 (5.13)	10.94 (4.75)	2.37 (1.29)
Small	7.55 (2.13)	11.31 (4.61)	11.34 (5.28)	12.04 (5.41)	4.49 (1.83)
Q2	9.38 (3.65)	11.75 (5.81)	12.03 (6.35)	11.81 (5.73)	2.43 (1.51)
Q3	8.76 (3.90)	11.03 (6.25)	10.19 (5.87)	10.18 (6.15)	1.42 (1.13)
Big	8.19 (4.08)	9.37 (4.85)	9.07 (5.20)	8.60 (4.24)	0.41 (0.44)
Big–Small	0.64 (0.22)	−1.94 (−0.96)	−2.27 (−1.24)	−3.43 (−1.75)	—
Panel B: Postformation ICC					
All	5.09 (16.85)	5.53 (18.85)	6.13 (21.60)	7.67 (22.82)	2.58 (15.17)
Small	8.67 (21.64)	8.14 (22.76)	7.89 (25.30)	7.97 (23.23)	−0.70 (−3.54)
Q2	7.06 (23.26)	6.29 (20.66)	5.82 (20.16)	5.73 (19.29)	−1.33 (−7.95)
Q3	6.47 (21.48)	5.64 (18.91)	5.09 (17.16)	4.58 (15.30)	−1.89 (−11.62)
Big	5.36 (19.40)	4.88 (15.88)	4.81 (14.35)	4.36 (11.71)	−1.00 (−4.75)
Big–Small	−3.31 (−10.47)	−3.27 (−17.93)	−3.08 (−15.64)	−3.62 (−15.67)	—

Note: We perform univariate sorts on *Ln(Amihud)* and triple-sorted portfolios in sequential order into quartiles. The first and second sorts are on *SIZE* and the third sort is on *Ln(Amihud)*. Variables are defined in Table 1. The breakpoints for all quartiles are from only NYSE stocks. For each *Ln(Amihud)* quartile, we average the four size quartiles of the second sort (see text for more details). The table then reports postformation statistics on the resulting 4 × 4 *SIZE*, *Ln(Amihud)* portfolios. We calculate postformation returns (in excess of the risk-free rate) 2 months after the formation period and postformation *ICC* (in excess of the long-term government bond yield) 1 month after the formation period. Monthly excess returns are annualized in Panel A. *t* statistics in parentheses are the standard ones for realized returns and calculated using the Newey and West (1987) correction with 36 lags for *ICC*. The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000. The sample period is 1977–2021.

3.2 | Fama-MacBeth regressions

In this section, we discuss results from FM regressions involving returns and *ICC* regressed on *Amihud* and other firm characteristics. For our base case, we use the following regression specification⁹:

$$\begin{aligned}
 R_{it} - R_{ft} = & a + b_1 \text{Ln}(Amihud_{it-2}) + b_2 \text{Ln}(SIZE_{it-2}) + b_3 \text{Ln}(B/M_{it-2}) \\
 & + b_4 \text{Return11}_{it-2} + b_5 \text{Return}_{it-1} + b_6 \text{Beta}_{it-2} + b_7 \text{GP/AT}_{it-2} \\
 & + b_8 \text{Ln}(1 + AG_{it-2}) + e_{it},
 \end{aligned}
 \tag{4}$$

where $(R_{it} - R_{ft})$ is the monthly excess returns in excess of monthly T-bill returns. We lag the independent variables (except the last-month return) by 2 months to minimize microstructure effects. We winsorize all independent variables at 0.5% and 99.5% levels. In addition, *Ln(Amihud)* is cross-sectionally standardized each month by subtracting the cross-sectional mean and dividing by the cross-sectional standard deviation.

The monthly *ICC* regression specification is as follows:

$$\begin{aligned} ICC_{it} = & a + b_1 Ln(Amihud_{it-1}) + b_2 Ln(SIZE_{it-1}) + b_3 Ln(B/M_{it-1}) \\ & + b_4 Return11_{it-2} + b_5 Return_{it-1} + b_6 Beta_{it-1} + b_7 GP/AT_{it-1} \\ & + b_8 Ln(1 + AG_{it-1}) + e_{it}, \end{aligned} \quad (5)$$

where ICC_{it} is the implied cost of capital (in excess of the 10-year treasury yield). All independent variables are lagged by only 1 month (except the 11-month return) since microstructure concerns are not an issue with *ICC*. We note that in principle, it is not necessary to lag illiquidity at all in *ICC* regressions, as the theoretical arguments suggest a contemporaneous positive relation between these variables. Nonetheless, to ensure that the inputs to *ICC* are available to the investor in real time when the relationship is estimated, and to minimize any endogeneity concerns, we lag illiquidity by 1 month. Not doing so gives substantially similar results. Similar to the return regression, all independent variables are winsorized and $Ln(Amihud)$ is cross-sectionally standardized.

The main results are presented in Table 4. Panel A presents results for all stocks in our sample, and Panel B presents results for small stocks and large stocks separately. Given the high negative correlation between $Ln(Amihud)$ and $Ln(SIZE)$, we estimate the cross-sectional regressions with and without size. In columns (2) and (3), where we estimate the regressions without $Ln(SIZE)$, the coefficient on $Ln(Amihud)$ has a uniformly positive sign in both return and *ICC* regressions. In the return regressions, however, the coefficient on $Ln(Amihud)$ is economically and statistically insignificant. In the *ICC* regressions, the coefficient on $Ln(Amihud)$ is positive and highly significant.

In columns (4) and (5), where we estimate the regressions with $Ln(SIZE)$, in the return regressions, the coefficient on $Ln(Amihud)$ is positive but statistically insignificant. In the *ICC* regressions, the $Ln(Amihud)$ coefficient flips sign and becomes significantly negative while the coefficient on $Ln(SIZE)$ remains significantly negative. This strongly suggests that $Ln(Amihud)$ is a proxy of $Ln(SIZE)$ and its positive relationship with *ICC* in the absence of $Ln(SIZE)$ is due to its high negative correlation with $Ln(SIZE)$. The negative relationship between $Ln(Amihud)$ and *ICC* in the presence of $Ln(SIZE)$ is also consistent with the portfolio results in Table 3. The coefficient of -0.383 for $Ln(Amihud)$ implies that a one standard deviation increase in $Ln(Amihud)$ leads to a decline of 0.38% in *ICC*.¹⁰

$Ln(B/M)$ is significantly positively related to both returns and *ICC* which is consistent with the risk interpretations in Fama and French (1992) and Berk et al. (1999). While *Beta* is not related to realized returns, it is significantly positively related to *ICC* suggesting that investors do require a higher cost of capital for higher beta stocks. Past price momentum, *Return11*, not surprisingly has a positive sign in the return regressions although it is negatively related to *ICC* which suggests that past winners have the low long-term cost of capital. This is inconsistent with risk-based interpretations of price momentum. While profitability (*GP/AT*), as expected, is positively related to monthly returns, it is significantly negatively related to *ICC* which suggests profitable firms tend to have the low long-term cost of capital. Asset growth is negatively related to both returns and *ICC*.

Panel B of Table 4 presents results for small and large stocks. The results in Table 3 suggest that illiquidity is perhaps priced (in monthly realized returns) only within small stocks. We explore this further in a multivariate regression setting. Stocks with market capitalization above the NYSE median market capitalization are classified as “Large” and those with market capitalization below the median are classified as “Small.” The results from the return regressions confirm that $Ln(Amihud)$ is related to monthly realized returns only in small stocks. The slope coefficients corresponding to $Ln(Amihud)$ are much bigger and highly significant in small stocks compared with large stocks – it is 0.194 (t statistic = 2.36) among small stocks and -0.019 (t statistic = -0.33) among large stocks. In contrast, the negative relationship between $Ln(Amihud)$ and *ICC* is significant among both large and small stocks (t statistics of -8.88 and -3.58 , respectively) although it is stronger among large stocks. This is consistent with the findings in Panel B of Table 3 which shows that the spread in *ICC* between High and Low *Amihud* illiquidity portfolios was larger in magnitude for larger size portfolios and more statistically significant.

The results thus far indicate that the *Amihud* illiquidity is not priced in *ICC*. It is premature to conclude from this however that illiquidity, in general, is not priced. Therefore, we turn next to alternate measures of illiquidity suggested in the literature and examine their relationship to *ICC*.

3.3 | Alternative measures of illiquidity

We consider the following five additional measures of illiquidity:

- ***Ln(Spread)*** is the logarithm of the percentage quoted bid-ask spread scaled by price.
- ***FHT*** is the Fong et al. (2017) measure of the proportion of zero days.

TABLE 4 Fama-MacBeth regressions for realized returns and ICC for Amihud illiquidity.

	Panel A: All stocks			
	Without size		With size	
	Return	ICC	Return	ICC
<i>Constant</i>	−0.497 (−1.50)	−4.592 (−2.32)	−0.660 (−1.17)	1.448 (0.67)
<i>Ln(Amihud)</i>	0.040 (0.90)	0.592 (9.03)	0.086 (0.97)	−0.383 (−2.99)
<i>Ln(SIZE)</i>	—	—	0.025 (0.42)	−0.705 (−6.36)
<i>Ln(B/M)</i>	0.156 (3.21)	1.435 (5.27)	0.158 (3.23)	1.293 (4.88)
<i>Return11</i>	0.680 (4.15)	−1.555 (−8.44)	0.669 (4.11)	−1.407 (−8.19)
<i>Return1</i>	−3.163 (−7.59)	−3.836 (−17.08)	−3.202 (−7.77)	−3.312 (−15.37)
<i>Beta</i>	0.056 (0.44)	1.657 (9.55)	0.059 (0.49)	1.479 (7.96)
<i>GP/AT</i>	0.574 (3.58)	−0.874 (−4.46)	0.577 (3.80)	−1.075 (−5.74)
<i>Ln(1 + AG)</i>	−0.410 (−3.40)	−0.804 (−2.17)	−0.416 (−3.49)	−0.953 (−2.71)
No. of stocks	1345	1042	1345	1042
Adj. R ²	6.5%	14.8%	6.9%	15.4%
	Panel B: Large and small stocks (with size)			
	Return		ICC	
	Large	Small	Large	Small
<i>Constant</i>	0.062 (0.09)	−1.825 (−2.86)	2.353 (0.78)	9.388 (3.34)
<i>Ln(Amihud)</i>	−0.019 (−0.33)	0.194 (2.36)	−0.940 (−8.88)	−0.458 (−3.58)
<i>Ln(SIZE)</i>	−0.072 (−1.15)	0.240 (3.02)	−0.801 (−7.55)	−1.668 (−8.10)
<i>Ln(B/M)</i>	0.146 (2.77)	0.160 (2.79)	1.309 (4.32)	1.018 (4.47)
<i>Return11</i>	0.581 (2.87)	0.677 (4.71)	−0.955 (−4.86)	−1.524 (−9.80)
<i>Return1</i>	−2.769 (−5.24)	−3.271 (−8.07)	−2.636 (−10.89)	−3.225 (−15.26)
<i>Beta</i>	0.045 (0.31)	0.069 (0.64)	1.718 (7.75)	1.266 (7.38)
<i>GP/AT</i>	0.488 (2.95)	0.651 (3.79)	−1.446 (−5.95)	−1.211 (−4.72)

TABLE 4 (Continued)

	Panel B: Large and small stocks (with size)			
	Return		ICC	
	Large	Small	Large	Small
$Ln(1 + AG)$	−0.283 (−2.07)	−0.458 (−2.89)	−1.099 (−2.62)	−0.847 (−2.63)
No. of stocks	660	684	617	428
Adj. R^2	10.5%	4.9%	15.6%	11.9%

Note: This table reports the results of Fama-MacBeth cross-sectional regressions for realized returns (in excess of the risk-free rate) and *ICC* (in excess of the yield on 10-year long-term government bonds) on Amihud illiquidity and various other firm characteristics. Variables are defined in Table 1. All independent variables are winsorized at 0.5% and 99.5% levels. $Ln(Amihud)$ is standardized into a Z score using cross-sectional mean and standard deviation each month. All variables are lagged by 2 months in return regressions and by 1 month in *ICC* regressions. *Return11* is lagged by 2 months, and *Return1* is lagged by 1 month in all regressions. Panel A presents results with size and without size. Panel B presents the results separately for small and large stocks with size. Stocks above the NYSE median market capitalization are defined as Large and the rest of the stocks are defined as Small. All coefficients are multiplied by 100. *t* statistics in parentheses are the standard ones for return regressions and are calculated using the Newey and West (1987) correction with 36 lags for *ICC* regressions. The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000. The sample period is 1977–2021.

- **Lambda** is the price impact coefficient of returns on the signed square root of dollar trading volume calculated using intraday transactions and then averaged over the month (obtained from WRDS).
- **GPIN** is the generalized PIN (probability of informed trading) measure of Duarte et al. (2020) that modifies the original *PIN* measure of Easley et al. (2002) which was estimated using daily signed buys and sells.¹¹ We follow the methodology in Duarte et al. (2020) to compute daily *GPIN* for NYSE stocks from 1993 to 2019.¹² We then compute the monthly average of these daily *GPIN*s and use the logistic transformation of this monthly average, $Ln[GPIN/(1 - GPIN)]$ as one of our measures of illiquidity. In the rest of the paper, when we refer to *GPIN*, we mean this logistic transformation.
- **CHL** is the Abdi and Rinaldo (2017) modified measure of the square root of the negative autocovariance of daily returns.
- **Composite** is a composite illiquidity measure based on the five measures listed above and the Amihud illiquidity measure (for a total of six illiquidity measures) constructed by computing the cross-sectional mean each month of the six standardized (Z scores) illiquidity measures.

Table A1 in the appendix reports correlations among the various illiquidity measures. The numbers above the diagonal are Spearman rank correlations and the numbers below the diagonal are Pearson correlations. Not surprisingly, all individual illiquidity measures are positively correlated with each other with rank correlations ranging from 26% to 66%. The one notable exception is *GPIN* which is weakly negatively correlated with every other illiquidity measure. As expected, $Ln(SIZE)$ is negatively correlated with all the illiquidity measures except with *GPIN*, where the correlation is weakly positive. As we will see below, *GPIN*'s cross-sectional pricing results also stand out from the rest of the illiquidity measures. *Composite* is highly negatively correlated with $Ln(SIZE)$ with a Pearson correlation of -0.755 and a rank correlation of -0.795 .

We now turn to examining the relationship between these six illiquidity measures and monthly returns and *ICC* using the same methodologies that we used to examine *Amihud* in Sections 3.1 and 3.2. Table 5 provides the results from bivariate portfolio sorts based on *SIZE* and illiquidity just as in Table 3 (64 *SIZE-SIZE-Illiquidity* portfolios collapsed to 16 *SIZE-Illiquidity* portfolios). In the interest of parsimony, we present only the (High – Low) spread between the high and low illiquidity portfolios for each *SIZE* quartile for each illiquidity measure.

Panel A of Table 5 presents the results for monthly returns and Panel B presents the results for *ICC*. As in Table 3, the first row presents the univariate results. The results for monthly returns show that only *GPIN* is positively priced in monthly returns (consistent with the findings in Easley et al., 2002) although the pricing is concentrated among small stocks. None of the other illiquidity measures are positively priced including the *Composite*. If anything, there is evidence of negative pricing for many of the measures among smaller stocks.

The results in Panel B of Table 5 show that *ICC* increases with illiquidity for every illiquidity measure other than *GPIN*. The results are highly significant for both univariate and bivariate portfolios. The positive premiums range between 1% and 3% and are higher for smaller stocks. The *GPIN* on the other hand exhibits a negative *ICC* premium which is the opposite of its relation to monthly returns. Overall, the results show that illiquidity is indeed positively

TABLE 5 Average return and ICC for portfolios sorted by size and other illiquidity measures.

SIZE	<i>Ln(Spread)</i>	<i>FHT</i>	<i>Lambda</i>	<i>GPIN</i>	<i>CHL</i>	Composite
Panel A: Postformation Return (High–low)						
All	−2.46 (−1.47)	−1.06 (−0.49)	0.93 (0.41)	3.09 (2.75)	−1.69 (−1.07)	0.15 (0.08)
Small	−5.12 (−3.03)	−4.31 (−3.42)	−2.47 (−0.69)	11.15 (1.58)	−4.17 (−2.19)	−5.94 (−3.01)
Q2	−5.34 (−2.99)	−2.05 (−1.10)	0.15 (0.07)	8.26 (2.79)	−3.89 (−2.94)	−3.87 (−2.42)
Q3	−0.32 (−0.21)	−1.97 (−1.20)	−0.13 (−0.06)	2.83 (0.93)	−1.44 (−1.10)	−1.31 (−1.01)
Big	−1.47 (−1.21)	0.38 (0.30)	−1.62 (−0.82)	1.07 (0.60)	−1.55 (−1.07)	−0.89 (−0.89)
Panel B: Postformation ICC (High – Low)						
All	2.98 (12.58)	3.12 (15.34)	2.94 (14.13)	−0.08 (−1.10)	3.00 (17.04)	3.61 (19.23)
Small	1.47 (5.20)	2.00 (11.29)	1.37 (3.45)	−1.99 (−4.23)	2.07 (9.66)	2.05 (8.83)
Q2	1.84 (7.00)	1.96 (10.52)	1.34 (4.83)	−0.33 (−1.17)	2.09 (12.89)	2.18 (11.18)
Q3	1.72 (8.12)	1.30 (9.82)	1.09 (3.61)	−0.07 (−0.59)	1.81 (11.92)	1.77 (11.63)
Big	1.27 (6.83)	0.50 (3.14)	1.03 (3.04)	−0.12 (−1.44)	1.36 (6.78)	1.23 (6.08)

Note: We consider univariate sorts on various illiquidity measures and triple-sorted portfolios in sequential order into quartiles. The first and second sorts are on *SIZE*, and the third sort is on illiquidity to construct sequentially sorted portfolios (as in Table 3). *Ln(Spread)* is the logarithm of the percentage quoted bid–ask spread. *FHT* is the Fong et al. (2017) measure of proportion of zero return days. *Lambda* is the price impact coefficient of returns on the signed square root of dollar trading volume calculated using intraday transactions and then averaged over the month. *GPIN* is the generalized PIN measure of Duarte et al. (2020) calculated as $\ln[PIN/(1 - PIN)]$. *CHL* is the Abdi and Rinaldo (2017) modified measure of the square root of the negative autocovariance of daily returns. The *Composite* measure is constructed as the cross-sectional mean each month of the six standardized (Z scores) illiquidity proxies (*Ln(Amihud)*, *Ln(Spread)*, *FHT*, *Lambda*, *GPIN*, and *CHL*). The breakpoints for all quintiles are from only NYSE stocks. For each illiquidity quartile, we average the four size quartiles from the second sort. The table then reports postformation statistics on the resulting High–Low portfolio for the univariate sort and the four size quartile portfolios. We calculate postformation returns (in excess of the risk-free rate) 2 months after the formation period and postformation ICC (in excess of the 10-year long-term government bond yield) 1 month after the formation period. Excess returns are annualized in Panel A. *t* statistics in parentheses are the standard ones for realized returns and are calculated using the Newey and West (1987) correction with 36 lags for ICC. The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000. The sample period is 1977–2021 for *Ln(Spread)*, *FHT*, *CHL*, and *Composite*; 1993–2020 for *Lambda*; and 1993–2019 for *GPIN*.

priced in long-term cost of capital. This validates the point made by Amihud and Mendelson (2008) that corporations should pay attention to the costs of illiquidity on the cost of capital. We now turn to confirm these findings in a multivariate setting.

Table 6 presents Fama–MacBeth cross-sectional regression results for the five new illiquidity measures and the composite illiquidity measure. The row entitled *XILLIQ* represents the results for the individual illiquidity measures. Panel A presents results for monthly returns. The results show that every illiquidity measure other than *GPIN* is significantly negatively related to monthly returns which are contrary to the predictions of illiquidity asset pricing models. *GPIN*, on the other hand, is significantly positively related to monthly returns. As we saw in Table A2, *GPIN* is also the only illiquidity measure that is not positively correlated with the other illiquidity measures. The results based on ICC (in Panel B of Table 6), which is a measure of long-term corporate cost of capital, present a completely different picture. Now every illiquidity measure other than *GPIN* is significantly positively priced in ICC suggesting that illiquidity does matter for the long-term corporate cost of capital. The composite illiquidity measure, *Composite*, is also significantly positively related to ICC. Overall, we conclude that illiquidity is indeed positively related to cost of capital/expected returns but to detect that relationship one needs to use a measure of long-term cost of capital such as ICC not short-term monthly returns.

TABLE 6 Fama-MacBeth regressions for realized returns and ICC for other illiquidity measures.

<i>XILLIQ</i>	<i>Ln(Spread)</i>	<i>FHT</i>	<i>Lambda</i>	<i>GPIN</i>	<i>CHL</i>	<i>Composite</i>
Panel A: Return						
<i>Constant</i>	−0.339 (−0.87)	−0.344 (−0.89)	0.062 (0.12)	0.609 (1.19)	−0.091 (−0.23)	0.115 (0.28)
<i>XILLIQ</i>	−0.113 (−3.31)	−0.085 (−4.11)	−0.076 (−2.24)	0.052 (2.45)	−0.139 (−3.60)	−0.174 (−4.53)
<i>Ln(SIZE)</i>	−0.037 (−1.31)	−0.045 (−1.74)	−0.033 (−0.96)	−0.053 (−1.44)	−0.071 (−2.67)	−0.106 (−3.77)
<i>Ln(B/M)</i>	0.157 (3.40)	0.166 (3.59)	0.103 (1.64)	0.053 (0.86)	0.153 (3.22)	0.152 (3.11)
<i>Return11</i>	0.690 (4.28)	0.663 (4.09)	0.385 (1.59)	0.267 (0.99)	0.690 (4.33)	0.687 (4.18)
<i>Return1</i>	−3.199 (−7.79)	−3.223 (−7.82)	−2.017 (−3.62)	−2.088 (−3.60)	−3.113 (−7.56)	−3.042 (−7.28)
<i>Beta</i>	0.117 (0.99)	0.077 (0.61)	0.100 (0.65)	0.055 (0.34)	0.086 (0.73)	0.093 (0.76)
<i>GP/AT</i>	0.533 (3.52)	0.543 (3.57)	0.543 (2.39)	0.059 (0.29)	0.595 (3.78)	0.565 (3.56)
<i>Ln(1 + AG)</i>	−0.431 (−3.62)	−0.449 (−3.76)	−0.410 (−2.32)	−0.318 (−1.96)	−0.437 (−3.63)	−0.437 (−3.61)
No. of stocks	1410	1410	1096	902	1309	1295
Adj. R^2	6.7%	6.6%	7.1%	8.1%	7.1%	7.0%
Panel B: ICC						
<i>Constant</i>	−1.427 (−0.77)	−1.012 (−0.48)	3.453 (4.09)	2.358 (2.18)	−1.180 (−0.55)	−3.162 (−1.53)
<i>XILLIQ</i>	0.668 (14.48)	0.357 (7.26)	0.371 (8.41)	0.001 (0.05)	0.633 (15.12)	0.892 (14.83)
<i>Ln(SIZE)</i>	−0.285 (−4.81)	−0.353 (−9.52)	−0.335 (−5.64)	−0.368 (−6.43)	−0.306 (−6.65)	−0.039 (−0.80)
<i>Ln(B/M)</i>	1.314 (4.88)	1.305 (4.97)	0.721 (5.78)	0.875 (5.00)	1.318 (4.91)	1.320 (4.90)
<i>Return11</i>	−1.335 (−8.30)	−1.361 (−7.94)	−1.294 (−5.84)	−1.501 (−6.62)	−1.368 (−8.52)	−1.331 (−8.17)
<i>Return1</i>	−3.755 (−17.47)	−3.691 (−16.50)	−3.549 (−13.88)	−3.346 (−14.56)	−3.551 (−16.87)	−3.954 (−17.44)
<i>Beta</i>	1.262 (8.74)	1.509 (9.24)	1.352 (11.27)	1.486 (10.35)	1.298 (7.83)	1.317 (9.14)
<i>GP/AT</i>	−0.996 (−5.46)	−1.044 (−5.66)	−1.349 (−7.16)	−0.847 (−3.12)	−1.133 (−5.87)	−1.055 (−5.60)
<i>Ln(1 + AG)</i>	−0.912 (−2.42)	−0.869 (−2.38)	−0.623 (−2.54)	−0.897 (−3.09)	−0.986 (−2.70)	−0.915 (−2.49)
No. of stocks	1048	1048	935	816	1027	1024
Adj. R^2	16.2%	15.7%	16.0%	15.9%	16.6%	16.6%

Note: This table reports the results of Fama-MacBeth cross-sectional regressions for realized returns (in excess of the risk-free rate) and ICC (in excess of the yield on long-term government bonds) on various illiquidity measures and various firm characteristics. Variables are defined in Tables 1 and 5. All independent variables are winsorized at 0.5% and 99.5% level. *XILLIQ* is a Z score standardized using cross-sectional mean and standard deviation each month. All variables are lagged by 2 months in return regressions and by 1 month in ICC regressions. *Return11* is lagged by 2 months, and *Return1* is lagged by 1 month in all regressions. All coefficients are multiplied by 100. *t* statistics in parentheses are the standard ones for return regressions and calculated using the Newey and West (1987) correction with 36 lags for ICC regressions. The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000. The sample period is 1977–2021 for *Ln(Spread)*, *FHT*, *CHL*, and *Composite*; 1993–2020 for *Lambda*; and 1993–2019 for *GPIN*. Note that *Composite* includes Amihud Illiquidity.

3.4 | Robustness tests

Now we turn to a series of robustness tests of our key findings. First, we estimate the regression specifications in Equations (4) and (5) in ranks, that is, where all the independent variables are converted to rank order with values between 0 and 1. The results are presented in Table A2 of the appendix. The results in columns (1) and (2) show that $Ln(Amihud)$ is marginally priced in monthly returns and negatively related to ICC —the negative relationship between $Ln(Amihud)$ and ICC is noticeably stronger with a t statistic of -7.63 . In the ICC regressions, the slope coefficients are annualized expected return premiums for each independent variable and are therefore comparable across them. The negative premium of -3.733% corresponding to $Ln(Amihud)$ is economically significant and comparable in magnitude to the premiums for $Ln(SIZE)$ (-6.042%) and $Ln(B/M)$ (2.729%). The results involving *Composite* (see columns 3 and 4) are a mirror image of those involving $Ln(Amihud)$. *Composite* is negatively related to monthly returns and significantly positively related to ICC consistent with the results in Panel B of Table 6.

Second, we examine the robustness of our findings using two variations of ICC based on Easton (2004) and Ohlson and Juettner-Nauroth (2005) which also rely on analyst earnings forecasts. These results are presented in Table A3 of the appendix. We find that the negative relation between $Ln(Amihud)$ and ICC is even stronger with the alternative $ICCs$ (see columns 1 and 2). For the Easton (2004) ICC measure, the t statistic corresponding to $Ln(Amihud)$ is -11.30 and for the Ohlson and Juettner-Nauroth (2005) ICC measure, the t statistic is -12.91 . The results based on *Composite* (see columns 3 and 4) show that the relation between *Composite* and the alternate $ICCs$ is positive although significant only for the Ohlson and Juettner-Nauroth (2005) ICC measure. Overall, the results show that our findings are not dependent on any one approach to estimating the ICC .

Third, in Table A4 of the appendix, we include additional factor betas beyond the market beta as controls. Specifically, we estimate betas using the five-factor Fama and French model, using the same approach as that used to estimate the market beta, and include them as controls. The results do not change appreciably. We find that ICC continues to be negatively related to $Ln(Amihud)$ and positively related to *Composite* while the monthly returns are positively related to $Ln(Amihud)$ and negatively related to *Composite*.

Finally, in Table A5 of the appendix, we consider the robustness of our findings by splitting the time-series into two equal halves and estimating the return and ICC regressions separately for each subsample. In the return regressions, $Ln(Amihud)$ is positive and marginally significant in the first half and negative and insignificant in the second half while *Composite* is negative and significant in the first half and negative and insignificant in the second half. In the ICC regressions, $Ln(Amihud)$ is negative and significant in the first half and negative and insignificant in the second half while *Composite* is positive and highly significant in both halves. This is important since this shows that the composite illiquidity measure is significantly positively priced in ICC in both subsamples.

4 | CROSS-SECTIONAL REGRESSIONS INVOLVING LIQUIDITY RISK

In this section, we investigate the role of liquidity risk in determining a firm's long-term cost of capital. A major branch of the literature on liquidity is the introduction of the role of liquidity risk. The idea is that fluctuations in illiquidity should be priced in addition to illiquidity itself. The two seminal contributions in this area are Acharya and Pedersen (2005) and Pástor and Stambaugh (2003). Following their approaches, we estimate two liquidity betas: *AP beta* is the liquidity beta calculated as in Acharya and Pedersen (2005), and *PS beta* is the liquidity beta calculated as in Pástor and Stambaugh (2003). The exact procedure is as follows:

AP beta: We form 25 illiquidity portfolios based on Acharya and Pedersen (2005) for the sample period from July 1962 to December 2021. Specifically, we sort NYSE and AMEX stocks with prices between \$5 and \$1000 based on their Amihud illiquidity over the prior year (with a minimum of 100 days of observations) into 25 portfolios. We form value-weighted portfolios at the end of December of each year with annual rebalancing and compute portfolio-level normalized illiquidity each month using prior month's market capitalization as weights. We also calculate illiquidity for a market portfolio of NYSE/AMEX stocks with prices between \$5 and \$1000 and at least 15 days of return and volume data each month and rebalance this portfolio monthly. Innovations in portfolio illiquidity, market illiquidity, and market returns are calculated using an AR(2) regression following Acharya and Pedersen (2005). From Equation (24) of their paper, we then calculate four illiquidity betas and a net illiquidity beta based on the four illiquidity betas. The net illiquidity beta (*AP beta*) is the measure of liquidity risk. The full-sample portfolio net illiquidity betas are then reassigned to individual stocks based on the portfolio to which the stock belongs at the end of each year.

PS beta: We form Pástor and Stambaugh (2003) 10 illiquidity portfolios for the sample period from December 1977 to December 2021. We first calculate liquidity betas for all stocks in regression with three Fama and French (1993) factors and a liquidity innovation factor. The regression uses monthly data over the past 5 years (with 60 observations) on a rolling basis. We then form decile portfolios for NYSE/AMEX stocks with prices between \$5 and \$1000 based on their liquidity betas. Portfolios are formed at the end of December of each year and rebalanced annually. The full-sample portfolio illiquidity betas are then reassigned to individual stocks based on the portfolio to which the stock belongs at the end of each year.

Table 7 reports results from cross-sectional regressions examining the relationship among returns, *ICC*, and the two liquidity betas. We use the composite illiquidity measure as a control in these regressions. The sample consists of NYSE/AMEX stocks and the time period is 1977 to 2021. Columns (1) and (2) present results for the *AP Beta*, and columns (3) and (4) present results for the *PS Beta*. The results with respect to liquidity risk are mixed. Neither risk measure is related to monthly returns. Although the *PS Beta* is significantly positively related to *ICC*, the *AP Beta* is significantly negatively related to *ICC*. Overall, the role of liquidity risk in the long-term cost of capital is undetermined. We want to emphasize that we are not implying that liquidity risk is irrelevant to a typical investor. Instead, we argue that we are not able to uncover evidence that liquidity risk should be of concern to firms in setting their long-term discount rates.¹³

TABLE 7 Fama-MacBeth regressions for realized returns and *ICC* using liquidity risk.

	AP Beta		PS Beta	
	Return	ICC	Return	ICC
<i>Constant</i>	−0.005 (−0.01)	4.197 (1.68)	0.048 (0.11)	−3.998 (−1.94)
<i>Composite</i>	−0.181 (−4.75)	0.724 (11.87)	−0.152 (−4.11)	0.848 (14.26)
<i>Ln(SIZE)</i>	−0.107 (−3.71)	−0.320 (−5.19)	−0.103 (−3.61)	−0.016 (−0.34)
<i>Ln(B/M)</i>	0.157 (3.13)	1.285 (4.79)	0.160 (3.09)	1.395 (5.14)
<i>Return11</i>	0.704 (3.87)	−1.359 (−8.30)	0.658 (3.43)	−1.449 (−8.87)
<i>Return1</i>	−3.097 (−7.40)	−3.828 (−17.81)	−3.347 (−7.94)	−3.943 (−18.93)
<i>Beta</i>	0.095 (0.78)	1.380 (8.75)	0.114 (0.89)	1.387 (8.15)
<i>GP/AT</i>	0.553 (3.47)	−1.135 (−6.01)	0.559 (3.48)	−0.912 (−5.35)
<i>Ln(1 + AG)</i>	−0.376 (−3.11)	−0.926 (−2.50)	−0.273 (−2.11)	−1.043 (−2.61)
<i>Liquidity Beta</i>	0.083 (0.31)	−4.519 (−11.14)	0.306 (0.41)	3.735 (2.63)
No. of stocks	1261	1011	1156	934
Adj. R^2	7.4%	17.1%	7.7%	17.2%

Note: This table reports the results of Fama-MacBeth cross-sectional regressions for realized returns (in excess of the risk-free rate) and *ICC* (in excess of the yield on long-term government bonds) on the composite illiquidity measure, various firm characteristics, and two proxies of liquidity betas. *AP Beta* is the liquidity beta calculated as in Acharya and Pedersen (2005), *PS Beta* is the liquidity beta is estimated as in Pástor and Stambaugh (2003). All other variables are defined in Tables 1 and 5. All independent variables are winsorized at 0.5% and 99.5% levels. All variables are lagged by 2 months in return regressions and by 1 month in *ICC* regressions. *Return11* is lagged by 2 months, and *Return1* is lagged by 1 month in all regressions. All coefficients are multiplied by 100. *t* statistics in parentheses are the standard ones for return regressions and are calculated using the Newey and West (1987) correction with 36 lags for *ICC* regressions. The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000. The sample period is 1977–2021.

5 | CROSS-SECTIONAL REGRESSIONS INVOLVING NASDAQ STOCKS

In this section, we examine the robustness of our cross-sectional findings in Section 3 involving NYSE/AMEX stocks using NASDAQ stocks. The filters employed for NASDAQ stocks are the same as those for NYSE/AMEX stocks, as described in Section 2. Further, it is known that trading volume is often overstated for NASDAQ stocks. Therefore, we use the procedure recommended by Anderson and Dyl (2005) to adjust for this overstatement.¹⁴ Since NASDAQ trading volume is not available prior to 1983, the sample period is 1977–2021 for *Ln(Spread)*, *FHT*, and *Composite*; 1983 to 2021 for *Ln(Amihud)* and *CHL*; and 1993–2020 for *Lambda*. *GPIN* is not available for NASDAQ stocks. The results are presented in Table 8 and are consistent with those for NYSE/AMEX stocks in Table 6. *Ln(Amihud)* is positively priced in monthly returns, whereas the rest of the illiquidity measures are negatively related to monthly returns. *Ln(Amihud)* is significantly negatively related to *ICC*, whereas the other illiquidity measures are significantly positively related to *ICC*. The results confirm our original finding that illiquidity is positively priced in long-term cost of capital and the cost of illiquidity should be of concern to corporations.

6 | ADDRESSING ENDOGENEITY CONCERNS

There remains a concern that endogeneity might affect inferences in our cross-sectional regressions of *ICC* on illiquidity. Endogeneity leads to the violation of the OLS assumption that the explanatory variables and the OLS error term are uncorrelated and may manifest itself in three ways: (a) omitted variables that are correlated with the explanatory variables, (b) simultaneity bias due to cost of capital and illiquidity being jointly and endogenously determined in equilibrium, and (c) measurement errors in the estimation of *ICC*, illiquidity, and other explanatory variables. In our base case *ICC* regressions, we attempted to mitigate this concern by lagging all independent variables by 1 month. But given the high serial dependence in *ICC*, this may not be enough to address endogeneity concerns. We follow two approaches: (1) address endogeneity concerns within the cross-sectional regression framework and (2) address endogeneity concerns using a Difference-in-Differences (DiD) methodology involving brokerage closures. We report below the outcome of our investigation to address these endogeneity concerns.

6.1 | Endogeneity tests within cross-sectional regressions

We try several things to address endogeneity concerns in cross-sectional regressions especially with respect to illiquidity: (a) an instrumental variable approach where we use lagged illiquidity and other explanatory variables as instruments for illiquidity, (b) using lagged *ICC* as a control to address the serial dependence in *ICC*, and (c) a panel regression approach with firm fixed effects and time fixed effects.

These results are presented in Table 9 for NYSE/AMEX stocks using *Composite* as our proxy for illiquidity. The *t* statistics are calculated using the Newey and West (1987) correction with 36 lags for Fama-MacBeth regressions and double clustering at firm and month levels for panel regressions. The instrumental variable regression is run using two-stage least squares. In the first stage, illiquidity at time *t* is regressed on illiquidity and all other explanatory variables at time *t* – 1. In the second stage, the fitted value is used in place of time *t* illiquidity to estimate the cross-sectional regression. Specification (1) is similar to the last column in Panel B of Table 6 (results are slightly different due to slight differences in sample selection). The results confirm that *Composite* is priced in *ICC*. Specification (2) provides the results of the instrumental variable regression. The results confirm the findings of Specification (1) and show that *Composite* is strongly priced in *ICC*. Specification (3) adds lagged *ICC* as a control to Specification (1) (note that this regression can be thought of as regressing changes in *ICC* on lagged *ICC*, *Composite*, and other independent variables). Not surprisingly, lagged *ICC* strongly predicts current *ICC*. More importantly, the composite is still significantly positively related to *ICC*. Specification (4) estimates Specification (3) as an instrumental variable regression and finds the same results.

Specifications (5) and (6) estimate panel regressions with firm and month fixed effects. The use of firm fixed effect is a way to address endogeneity concerns if the endogeneity (affecting *ICC* and illiquidity) is firm-specific but time invariant. Specification (5) estimates the panel regression with *t* – 1 composite illiquidity, and Specification (6) estimates the panel regression with instrumented illiquidity. The results show that in both regressions *Composite* is highly significantly priced in *ICC*.

TABLE 8 Fama-MacBeth regressions for realized returns and ICC for NASDAQ stocks.

<i>XILLIQ</i>	<i>Ln(Amihud)</i>	<i>Ln(spread)</i>	<i>FHT</i>	<i>Lambda</i>	<i>CHL</i>	<i>Composite</i>
Panel A: Return						
<i>Constant</i>	−2.443 (−3.24)	−1.020 (−1.96)	−0.931 (−1.99)	−0.591 (−0.89)	−0.943 (−1.75)	−0.829 (−1.48)
<i>XILLIQ</i>	0.248 (2.32)	−0.050 (−0.91)	−0.109 (−4.76)	−0.049 (−1.69)	−0.198 (−5.05)	−0.151 (−3.27)
<i>Ln(SIZE)</i>	0.276 (3.73)	0.024 (0.61)	0.009 (0.30)	0.067 (1.52)	0.038 (1.03)	0.020 (0.50)
<i>Ln(B/M)</i>	0.234 (3.67)	0.227 (4.03)	0.223 (3.94)	0.134 (1.67)	0.223 (3.47)	0.223 (3.39)
<i>Return11</i>	0.612 (5.66)	0.732 (7.88)	0.733 (7.84)	0.286 (2.27)	0.592 (5.48)	0.633 (5.72)
<i>Return1</i>	−3.208 (−8.70)	−3.607 (−11.06)	−3.625 (−11.07)	−2.756 (−6.01)	−3.161 (−8.29)	−3.175 (−8.34)
<i>Beta</i>	0.082 (0.73)	0.067 (0.61)	0.073 (0.64)	0.083 (0.57)	0.064 (0.52)	0.024 (0.20)
<i>GP/AT</i>	0.765 (5.34)	0.727 (5.82)	0.729 (5.84)	0.782 (4.27)	0.663 (4.52)	0.697 (4.74)
<i>Ln(1 + AG)</i>	−0.612 (−5.77)	−0.637 (−6.00)	−0.653 (−6.13)	−0.598 (−4.33)	−0.731 (−6.49)	−0.696 (−6.34)
No. of stocks	1233	1341	1340	1296	1200	1175
Adj. R^2	5.5%	5.4%	5.3%	5.5%	5.4%	5.3%
Panel B: ICC						
<i>Constant</i>	6.293 (7.27)	2.599 (2.70)	2.554 (2.93)	5.026 (6.81)	3.098 (4.17)	2.264 (3.10)
<i>XILLIQ</i>	−0.549 (−3.24)	0.148 (1.78)	0.282 (10.50)	0.114 (4.01)	0.360 (8.72)	0.457 (6.74)
<i>Ln(SIZE)</i>	−1.201 (−7.99)	−0.759 (−10.65)	−0.743 (−17.00)	−0.771 (−10.76)	−0.716 (−11.70)	−0.585 (−6.89)
<i>Ln(B/M)</i>	0.930 (10.16)	1.029 (8.38)	1.031 (8.53)	0.791 (8.54)	0.965 (9.55)	0.954 (9.39)
<i>Return11</i>	−0.700 (−5.27)	−0.849 (−6.80)	−0.823 (−6.89)	−0.679 (−5.54)	−0.785 (−6.08)	−0.804 (−6.24)
<i>Return1</i>	−2.367 (−13.63)	−2.781 (−15.56)	−2.891 (−15.13)	−2.655 (−12.88)	−2.699 (−16.31)	−2.864 (−17.39)
<i>Beta</i>	1.000 (6.94)	1.139 (8.16)	1.117 (7.75)	1.021 (6.03)	1.086 (7.14)	1.127 (7.40)
<i>GP/AT</i>	−0.184 (−0.63)	−0.178 (−0.52)	−0.214 (−0.61)	0.349 (1.16)	−0.057 (−0.19)	0.018 (0.06)
<i>Ln(1 + AG)</i>	0.355 (1.99)	0.451 (2.50)	0.436 (2.39)	0.505 (2.79)	0.492 (2.89)	0.545 (3.13)
No. of stocks	849	791	794	902	830	823
Adj. R^2	15.7%	15.2%	15.4%	15.6%	15.7%	15.7%

Note: This table reports the results of Fama-MacBeth cross-sectional regressions for realized returns (in excess of the risk-free rate) and ICC (in excess of the yield on 10-year long-term government bonds) on various illiquidity measures and various firm characteristics as in Tables 4 and 6 except that the sample consists only of NASDAQ stocks with prices between \$5 and \$1000. The sample period is 1977–2021 for *Ln(Spread)*, *FHT*, and *Composite*; 1983–2021 for *Ln(Amihud)* and *CHL*; and 1993–2020 for *Lambda*. Note that the *Composite* does not include *GPIN* as it is available only for NYSE stocks.

TABLE 9 Cross-sectional regressions addressing endogeneity.

	FM				Panel	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Constant</i>	−3.213 (−1.56)	−4.743 (−2.35)	−0.448 (−1.56)	−0.610 (−2.12)	—	—
<i>Composite</i>	0.883 (15.09)	—	0.100 (8.64)	—	0.542 (16.28)	—
<i>InstComposite</i>	—	1.390 (16.08)	—	0.156 (9.49)	—	1.110 (16.54)
<i>Ln(SIZE)</i>	−0.033 (−0.67)	0.200 (3.04)	−0.006 (−0.81)	0.018 (1.99)	−1.008 (−15.26)	−0.730 (−10.39)
<i>Ln(B/M)</i>	1.318 (4.89)	1.310 (4.84)	0.184 (4.94)	0.184 (4.89)	0.807 (11.18)	0.776 (10.91)
<i>Return11</i>	−1.377 (−8.31)	−1.299 (−8.13)	−0.088 (−2.99)	−0.077 (−2.67)	−0.955 (−12.97)	−0.919 (−12.80)
<i>Return1</i>	−3.974 (−18.44)	−3.220 (−15.92)	−2.448 (−17.50)	−2.355 (−18.84)	−3.249 (−20.68)	−2.808 (−18.66)
<i>Beta</i>	1.322 (8.83)	1.191 (8.45)	0.168 (5.70)	0.154 (5.46)	0.223 (3.00)	0.129 (1.75)
<i>GP/AT</i>	−1.083 (−5.87)	−1.084 (−5.92)	−0.113 (−4.41)	−0.116 (−4.47)	−4.443 (−11.19)	−4.324 (−10.94)
<i>Ln(1 + AG)</i>	−0.919 (−2.39)	−0.907 (−2.35)	−0.039 (−1.05)	−0.039 (−1.02)	−0.667 (−5.93)	−0.631 (−5.66)
<i>ICC</i>	—	—	0.863 (165.82)	0.862 (163.38)	—	—
Adj. R^2	16.8%	16.8%	78.9%	78.9%	8.6%	8.3%

Note: This table reports results of regressions for *ICC* (in excess of the yield on 10-year long-term government bonds) on the composite illiquidity measure and various firm characteristics. Columns (1) through (4) report results from cross-sectional Fama-MacBeth regressions, whereas columns (5) and (6) report results from panel regressions (with firm and month fixed effects). Specification (1) is similar to the last column in Panel B of Table 6 (slight differences are due to slight differences in sample selection). The instrumented liquidity measure, *InstComposite*, is calculated by running a first-stage regression of composite liquidity measure on lagged composite liquidity measure and the other controls (except lagged *ICC*) listed in this table. All variables, except *Return11* are lagged by 1 month. *Return11* is lagged by 2 months. Lagged *ICC* (lagged by 1 month) is included as an additional control in columns (3) and (4). *t* statistics in parentheses are calculated using the Newey and West (1987) correction with 36 lags for FM regressions and using double clustering at firm and month level for panel regressions. Variables are defined in Tables 1 and 5. The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000. The sample period is 1977–2021.

6.2 | Difference-in-Differences (DiD) around brokerage closures

In this section, we focus on the effects of changes in liquidity of individual stocks due to exogenous reasons. For this experiment, we follow Kelly and Ljungqvist (2012) and construct a sample of stocks whose sell-side analyst coverage declined following brokerage closures or brokerage mergers that occurred between the first quarter of 2000 and the first quarter of 2008.¹⁵ For each brokerage closure, we identify stocks that were covered by the broker before the closure ended the coverage. For mergers, we identify stocks that had a sell-side earnings estimate by both brokers in the quarter before the event but by only one of the two brokers in the quarter after the event. This sample constitutes the sample of treated (not necessarily unique) stocks numbering 2563. For every treated stock, we choose five control stocks at random in the same size and book-to-market quintile as the treated stock in the quarter preceding the event; subject to the condition that the control firms were themselves not subject to coverage termination in the 1 year around the event.

We calculate *Ln(Amihud)*, *Composite*, and *ICC* for each treated stock in the month immediately before and the month immediately after the event. We choose this window to avoid the confounding effects of any other exogenous firm-specific events that might affect a stock's liquidity. The window also allows us to compute changes in *ICC* based on the *ICCs* computed immediately before and immediately after the brokerage closure thus reflecting any changes in consensus earning

estimates resulting from the brokerage closures. Thus, the window is likely to provide the cleanest test of the effect of brokerage closures on the cost of capital. We also average $Ln(Amihud)$, $Composite$, and ICC across the five control stocks corresponding to each treated stock to obtain one number for that control group (for the same 1-month window).

Table 10 reports the cross-sectional averages of the above statistics before and after the event for both treated and control firms as well as the double difference. Since brokerage closures are clustered in time, multiple brokerages can stop covering a stock at the same time, so that the same stock can show up as a treated stock multiple times within a short time period. To control for this clustering effect, we follow Kelly and Ljungqvist (2012) in calculating the standard error of the DiD with a block bootstrap simulation procedure with a block length of 100 and 10,000 repetitions. We report these statistics for the entire sample as well as for subsamples grouped based on the number of sell-side analysts covering the treated firm in the month preceding the event.

TABLE 10 Brokerage coverage terminations, illiquidity, and ICC .

No. of analysts	Treated		Control		DiD		
	Before	After	Before	After	Coeff	SE	N
Panel A: $Ln(Amihud)$							
All	−6.05	−6.02	−5.41	−5.44	0.061	0.012	2564
≤5	−2.71	−2.64	−2.14	−2.14	0.068	0.023	414
5–10	−5.04	−5.02	−4.48	−4.50	0.042	0.015	578
10–15	−6.36	−6.31	−5.77	−5.80	0.075	0.021	479
15–20	−7.14	−7.13	−6.66	−6.70	0.050	0.014	434
>20	−8.31	−8.28	−7.34	−7.39	0.070	0.014	644
Panel B: $Composite$							
All	−0.66	−0.63	−0.57	−0.55	0.023	0.010	2452
≤5	−0.04	−0.02	0.12	0.11	0.031	0.026	401
5–10	−0.54	−0.54	−0.43	−0.42	−0.017	0.014	566
10–15	−0.74	−0.71	−0.67	−0.66	0.018	0.012	465
15–20	−0.82	−0.78	−0.81	−0.79	0.024	0.015	424
>20	−1.08	−0.98	−0.94	−0.90	0.058	0.020	583
Panel C: ICC							
All	10.50	10.58	10.13	10.23	−0.010	0.120	1999
≤5	12.21	12.14	12.21	12.03	0.116	0.213	208
5–10	11.23	11.30	10.91	11.08	−0.108	0.178	444
10–15	11.41	11.71	10.43	10.63	0.103	0.299	405
15–20	9.77	9.70	9.80	9.82	−0.094	0.126	371
>20	9.12	9.23	8.78	8.90	−0.006	0.072	571
Panel D: $Ln(SIZE)$							
All	7.64	7.58	7.37	7.33	−0.020	0.014	2557
≤5	5.33	5.26	5.03	5.00	−0.047	0.012	417
5–10	6.72	6.65	6.57	6.52	−0.019	0.016	574
10–15	7.69	7.62	7.61	7.56	−0.017	0.025	473
15–20	8.44	8.39	8.31	8.28	−0.018	0.024	434
>20	9.51	9.45	8.91	8.85	−0.007	0.021	640

Note: This table reports results of a difference-in-differences (DiD) test around analyst coverage terminations by brokerage firms. The sample construction is described in the text. Control firms are chosen by selecting five stocks at random in the same size and book-to-market quintiles as the treated firm in the quarter preceding the event subject to the condition that the control firms were themselves not subject to coverage termination in the 1 year around the event. For each treated firm, we calculate the statistic of interest in the first month before and after the event. We repeat this calculation for each control firm and take the average across all control firms. The table reports the cross-sectional averages of these statistics before and after the event for both treated and control firms as well as the double difference (reported under the column “Coeff”). We calculate the standard error (SE) of the DiD difference with a block bootstrap of block length 100 and 10,000 repetitions. The number of observations is reported in column “N.” We report these statistics for the entire sample as well as the sample broken up based on the number of analysts for the treated firm in the month preceding the event.

The results in Panel A of Table 10 show that $Ln(Amihud)$ increases significantly (meaning liquidity declines) following brokerage closures in the overall sample and in every subsample by the number of analysts. Specifically, in the overall sample, $Ln(Amihud)$ increases by six standard deviations, whereas in the subsamples, it increases by three to seven standard deviations. These results confirm the findings of Kelly and Ljungqvist (2012). Yet, the composite illiquidity measure, *Composite*, which combines all measures of illiquidity does not increase as much (see Panel B of Table 10). In the overall sample, it increases by two standard deviations but in all subsamples by the number of analysts but one the increase is insignificant. This calls into question whether the decline in liquidity around brokerage closures is material.

The point estimates of *ICC* (see Panel C of Table 10) in fact decline a bit but in the subsamples by number of analysts, they increase in two subsamples and decline in other subsamples with no obvious pattern to be discerned. To put this finding in context, Kelly and Ljungqvist (2012) report that in the days after a stock loses coverage, its price drops on average by about 1.0% (adjusting for market movements). Similarly, we find that in our sample of 2563 firms, prices decline by about 0.8% (net of control firm price changes) during the event month. If consensus earnings estimates do not change appreciably from the month before to the month after the event, a decline in price should mechanically lead to an increase in *ICC*. We do not see such uniform increases in *ICC*, instead we see a mix of increases and decreases across the various subsamples. Although the Amihud illiquidity increases significantly after these events, the composite illiquidity does not change appreciably, and the *ICC* follows suit.

Panel D of Table 10 provides the DiD results for market capitalization. This is because market capitalization is negatively related to *ICC* in cross-sectional regressions (Lee et al., 2009, and our Table 4) and might possibly be an (inverse) illiquidity proxy. We find that while market capitalization declines after brokerage closures, the changes are economically modest except for firms covered by five analysts and are less and statistically insignificant in all cases but one. This is consistent with the modest changes in the composite illiquidity observed in Panel B.

7 | SUMMARY AND CONCLUDING REMARKS

A large body of literature finds evidence that high illiquidity implies high monthly realized returns. This finding is interpreted as supporting the notion that investors demand higher returns for more illiquid firms. Using standard U.S. data, we test if illiquidity is related to estimates of firms' costs of capital (*ICC*), which correspond to long-term internal rates of return (in excess of a benchmark 10-year Treasury yield) on expected cash flows. We are unable to uncover cross-sectional evidence that the Amihud (2002) measure of illiquidity is positively related to costs of capital, after controlling for other known determinants of *ICC*. A DiD analysis around exogenous brokerage closures shows a decrease in the Amihud illiquidity measure (as in Kelly & Ljungqvist, 2012) but no accompanying increase in *ICC*.

We consider an arsenal of alternative, and more economically intuitive, proxies for illiquidity, liquidity risk, and information risk. While several measures of illiquidity other than the Amihud measure are indeed priced in *ICC*, we find no evidence that information or liquidity risk is priced. Our findings shed specific light on precisely which illiquidity measures command a premium in firms' required returns.

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ENDNOTES

- ¹ See also Datar et al. (1998). Pástor and Stambaugh (2003), Acharya and Pedersen (2005), Kamara et al. (2008), Korajczyk and Sadka (2008), and Lou and Sadka (2011) extend the liquidity pricing context to systematic liquidity and liquidity risk pricing. Ben-Rephael et al. (2015) analyze the trend in the liquidity premium and show that it has diminished over the years.
- ² Amihud and Levi (2019) use this idea to investigate the link between illiquidity and real investment.
- ³ See Fama and French (1997) who find that the cost of capital estimates based on average realized returns are “unavoidably imprecise.” Elton (1999) makes a similar point in his AFA presidential address.
- ⁴ We eschew decimalization, owing to the controversy surrounding the impact of this event on trading costs (Chakravarty et al., 2005 and Eaton et al., 2021).
- ⁵ Berk (1993) proposes that market capitalization can be priced because it measures omitted risk-based variables. He notes that high values of such risk proxies would depress the price, implying lower market capitalization (with higher *ICC* and higher required returns).
- ⁶ Also see Pástor et al. (2008) and Lee et al. (2009).

- ⁷ We use the term “dividends” interchangeably with free cash flows to equity (*FCFE*) to describe all cash flows available to equity.
- ⁸ Results from portfolio sorts are less influenced by outliers since they are analogous to discrete ranks. Using continuous ranks (from 0 to 1) of independent variables in cross-sectional regressions is the equivalent of portfolio sorts since the slope coefficients can be interpreted as returns on zero-investment long-short portfolios constructed on that variable in a multivariate setting. Multivariate portfolio sorts, on the other hand, can better capture any non-linearity in the relationship between the variable of interest and expected returns.
- ⁹ Estimating the regressions without the market beta does not significantly change any of our findings. We add market beta because it is an important risk control for *ICC* and is positively correlated with *ICC* (see Table 2).
- ¹⁰ We use the maximum number of stocks available for each estimation. However, using the same sample across the *ICC* and realized returns regressions does not materially alter the conclusions. Results are available on request. We also examine the time-series of FM coefficients on $\ln(\text{Amihud})$ and find that outliers do not drive our result of a negative relation between illiquidity and *ICC*. The coefficient remains below zero for sustained periods of time, though there also are episodes of positivity.
- ¹¹ The key contribution of Easley et al. (2002) is to measure the probability of informed trading (or information risk) directly using a structural model and to show that this measure commands a positive premium in the cross-section of monthly stock returns. The logic is that informed trading distorts the portfolios of uninformed traders, thus inducing them to demand a return premium. This return premium could also extend to a premium in firms' cost of capital or *ICC* (Easley & O'Hara, 2004).
- ¹² We thank the authors for providing code and data on <https://edwinhu.github.io/pin/>.
- ¹³ Keloharju et al. (2021) show that long-term realized returns do not incorporate premia associated with a large number of “return anomalies” that emerge in short-term (monthly) realized returns. We instead focus on the relation between illiquidity and *ICC*, where the latter is a proxy for the actual cost of capital for long-term corporate projects. Our design sheds light on whether publicly traded corporations should account for illiquidity in setting their discount rates.
- ¹⁴ We correct for double counting of volume in NASDAQ by following Anderson and Dyl (2005). Specifically, for the periods prior to February 2001, February to December 2001, and calendar years 2002–2004, we divide NASDAQ volume by 2.0, 1.8, and 1.6, respectively.
- ¹⁵ We thank Feng Jiang for his help in constructing these data.
- ¹⁶ For instance, the average used for the *ICC* calculations in calendar year 1977 is the arithmetic average of annual nominal GDP growth rates from 1930 to 1976, the average used for 1978 is the arithmetic average of annual nominal GDP growth rates from 1930 to 1977, and so on.
- ¹⁷ If the fiscal year earnings are negative, then we scale dividends by median FY_1 earnings forecasts.
- ¹⁸ To gauge the impact of share repurchases and new equity issuances, we reestimate the payout ratio by incorporating share repurchases net of new equity issuances. Our results are robust to *ICC* estimated using this alternate payout ratio.
- ¹⁹ Note that the use of the no-growth perpetuity formula does not imply that earnings or cash flows do not grow after period $t + T$. Rather, it simply means that any new investments after year $t + T$ will earn zero economic profits. In other words, any growth in earnings or cash flows after year T is value irrelevant.
- ²⁰ Lee et al. (2009) and Li et al. (2013) test the robustness of *ICC* estimates by using different values of T ($T = 10$ or $T = 20$) and report that their cross-sectional and time-series results are robust to these alternative time horizons.
- ²¹ See also Balakrishnan et al. (2021) who construct cost of equity (CoE) estimates directly from analysts' CoE estimates which necessarily limits their cross-sectional sample to a few hundred firms per month on average. We have also estimated alternate *ICCs* based on Easton (2004) and Ohlson and Juettner-Nauroth (2005). We discuss the results using these alternate *ICCs* in Section 3.2.

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APPENDIX A

Details of the ICC Estimation

We use a three-stage approach to forecasting earnings up to year $t + T$. In the first stage, earnings forecast for year $t + 1$ is based on median security analyst earnings forecasts, FY_1 and FY_2 , for the next two fiscal years, where $FY_1 > 0$ and $FY_2 > 0$. We use a weighted average of the FY_1 and FY_2 to construct a 12-month ahead earnings forecast $FE_1 = w \times FY_1 + (1 - w) \times FY_2$, where w is the number of months to the next fiscal year-end divided by 12. In the second stage, we compute the implied growth rate, $g_2 = FY_2 / FY_1 - 1$, and use it to forecast the 2-year-ahead earnings forecast, $FE_2 = FE_1 \times (1 + g_2)$. This ensures that the forecasts for the next 2 years are always 12 months and 24 months ahead from the current month. The implied growth rate g_2 is winsorized to be between 1% and 75%. In the third stage, earnings from year $t + 3$ to year $t + T + 1$ are estimated by assuming that the year $t + 2$ earnings growth rate g_2 mean-reverts exponentially to its steady-state value by year $t + T + 2$. The steady-state growth rate in year $t + T + 2$ is assumed to be the long-run nominal GDP growth rate, g , which is computed as an expanding average of annual nominal GDP growth rates starting in the calendar year 1930 and ending in the prior calendar year.¹⁶ Thus, earnings growth rates and earnings forecasts for years $t + 3$ to $t + T + 1$ ($k = 3, \dots, T + 1$) are computed as follows:

$$g_{t+k} = g_{t+k-1} \times \exp[\log(g/g_2)/T],$$

$$FE_{t+k} = FE_{t+k-1} \times (1 + g_{t+k}). \quad (A1)$$

The exponential rate of mean reversion is the same as linear interpolation in logs and provides a more rapid rate of mean reversion for very high growth rates. This is consistent with the evidence in Nissim and Penman (2001) and Chan et al. (2003).

We forecast plowback rates also using a two-stage approach. In the first stage, plowback rate for year $t + 1$ is estimated as one minus the most recent year's dividend payout ratio. The dividend payout ratio is calculated as the ratio of the most recent fiscal year dividends to the most recent fiscal year (positive) earnings.¹⁷ We exclude share repurchases and new equity issues due to the difficulty in forecasting their recurrence in future periods.¹⁸ We winsorize payout ratios to be between zero and one. In the second stage, we assume that the plowback rate in year $t + 1$, b_1 reverts linearly to a steady-state value by year $t + T + 1$ computed from the sustainable growth rate formula which assumes that the product of the return on new investments and the plowback rate $ROE \times b$ is equal to the steady-state growth in earnings g . Because competition will drive the return on these investments down to the cost of equity, we further assume that in the steady state, ROE (return on new investments) equals r_e . Substituting ROE with cost of equity r_e in the sustainable growth rate formula gives us the steady-state value for the plowback rate, g/r_e . We compute the intermediate plowback rates from $t + 2$ to $t + T$ ($k = 2, \dots, T$) using linear interpolation:

$$b_{t+k} = b_{t+k-1} - (b_1 - b)/T. \quad (A2)$$

The terminal value TV is the present value of perpetuity equal to the year $t + T + 1$ earnings forecast divided by the cost of equity¹⁹:

$$TV_{t+T} = FE_{t+T+1} / r_e, \quad (A3)$$

It is easy to show that the constant growth model for TV simplifies to Equation (A3) when ROE equals r_e . Combining Equations (1), (2), and (A1)–(A3) provides the following empirically implementable finite horizon model:

$$P_t = \sum_{k=1}^T \frac{FE_{t+k} \times (1 - b_{t+k})}{(1 + r_e)^k} + \frac{FE_{t+T+1}}{r_e(1 + r_e)^T}. \quad (A4)$$

Following Li et al. (2013), we use a 15-year horizon ($T = 15$) to implement the model in Equation (A4) and compute r_e as the rate of return that equates the present value of free cash flows to the current stock price.²⁰ The resulting r_e is the firm-level ICC measure used in our empirical analysis. We compute ICC for each month from January 1977 to December 2021 for all firms with available positive fiscal year 1 and fiscal year 2 consensus earnings forecasts ($FY_1 > 0$, $FY_2 > 0$) and the other data requirements described earlier.²¹

TABLE A1 Cross-sectional correlations among various illiquidity measures.

	<i>Ln(Spread)</i>	<i>FHT</i>	<i>Lambda</i>	<i>CHL</i>	<i>GPIN</i>	<i>Ln(Amihud)</i>	<i>Composite</i>	<i>Ln(SIZE)</i>
<i>Ln(Spread)</i>		0.350	0.534	0.496	−0.092	0.486	0.784	−0.510
<i>FHT</i>	0.390		0.257	0.291	−0.009	0.423	0.586	−0.456
<i>Lambda</i>	0.210	0.231		0.394	−0.045	0.658	0.680	−0.656
<i>CHL</i>	0.495	0.391	0.246		−0.018	0.418	0.717	−0.449
<i>GPIN</i>	−0.071	−0.002	−0.004	−0.008		−0.124	0.148	0.116
<i>Ln(Amihud)</i>	0.499	0.431	0.324	0.435	−0.068		0.768	−0.947
<i>Composite</i>	0.760	0.693	0.527	0.742	0.224	0.784		−0.795
<i>Ln(SIZE)</i>	−0.498	−0.446	−0.312	−0.452	0.056	−0.940	−0.755	

Note: This table provides a time-series average of cross-sectional correlations among various illiquidity measures. Variables are defined in Tables 1 and 5. The upper part of the matrix (in italics) contains rank correlations. The lower part of the matrix contains correlations in levels. The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000. The sample period is 1977–2021 for *Ln(Spread)*, *FHT*, *CHL*, and *Ln(Amihud)*; 1993–2020 for *Lambda*; and 1993–2019 for *GPIN*.

TABLE A2 Fama-MacBeth Regressions for realized returns and ICC using ranks.

<i>XILLIQ</i>	<i>Ln(Amihud)</i>		<i>Composite</i>	
	Return	ICC	Return	ICC
<i>Constant</i>	0.019 (0.04)	10.681 (14.26)	0.875 (3.63)	4.849 (12.10)
<i>XILLIQ</i>	0.614 (1.88)	−3.733 (−7.63)	−0.268 (−1.82)	2.533 (13.24)
<i>Ln(SIZE)</i>	0.448 (1.16)	−6.042 (−10.79)	−0.371 (−2.26)	−0.513 (−1.98)
<i>Ln(B/M)</i>	0.455 (3.46)	2.729 (5.45)	0.415 (3.19)	2.897 (5.86)
<i>Return11</i>	0.859 (4.58)	−1.768 (−9.38)	0.918 (4.82)	−1.856 (−11.25)
<i>Return1</i>	−1.124 (−7.89)	−0.909 (−14.45)	−1.114 (−7.80)	−1.114 (−16.36)
<i>Beta</i>	0.093 (0.46)	2.541 (8.47)	0.083 (0.43)	2.309 (9.43)
<i>GP/AT</i>	0.519 (4.51)	−1.115 (−6.47)	0.484 (4.06)	−0.933 (−5.94)
<i>Ln(1 + AG)</i>	−0.156 (−2.04)	−1.141 (−4.87)	−0.158 (−2.05)	−1.120 (−4.81)
No. of stocks	1345	1042	1295	1024
Adj. R^2	6.7%	16.9%	6.8%	17.1%

Note: This table reports the results of Fama-MacBeth cross-sectional regressions for realized returns (in excess of the risk-free rate) and ICC (in excess of the yield on 10-year long-term government bonds) on Amihud illiquidity or the Composite illiquidity and various firm characteristics as in Tables 4 and 6 except that we use ranks of all independent variables (cross-sectionally assigned each month and scaled to lie between 0 and 1). Variables are defined in Tables 1 and 5. All variables are lagged by 2 months in return regressions and by 1 month in ICC regressions. *Return11* is lagged by 2 months and *Return1* is lagged by 1 month in all regressions. All coefficients are multiplied by 100. t statistics in parentheses are the standard ones for return regressions and are calculated using the Newey and West (1987) correction with 36 lags for ICC regressions. The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000. The sample period is 1977–2021.

TABLE A3 Fama-MacBeth regressions for realized returns and alternative measures of ICC.

	<i>Ln(Amihud)</i>		<i>Composite</i>	
	Easton (2004)	Ohlson and Juettner-Nauroth (2005)	Easton (2004)	Ohlson and Juettner-Nauroth (2005)
<i>Constant</i>	5.321 (1.21)	5.535 (1.70)	−9.843 (−2.93)	−6.178 (−2.23)
<i>XILLIQ</i>	−3.593 (−11.30)	−2.277 (−12.91)	0.179 (1.18)	0.847 (6.02)
<i>Ln(SIZE)</i>	−1.208 (−3.45)	−1.386 (−7.15)	0.798 (3.54)	0.210 (1.48)
<i>Ln(B/M)</i>	0.953 (3.97)	1.039 (4.95)	1.083 (4.21)	1.143 (5.14)
<i>Return11</i>	−1.673 (−5.81)	−1.674 (−6.92)	−2.402 (−10.33)	−2.043 (−9.43)
<i>Return1</i>	−0.682 (−1.66)	−0.367 (−1.19)	−2.840 (−7.64)	−2.156 (−6.38)
<i>Beta</i>	3.216 (10.27)	3.066 (11.42)	3.516 (13.92)	3.124 (15.11)
<i>GP/AT</i>	−0.449 (−1.35)	−1.207 (−3.68)	0.066 (0.22)	−0.860 (−2.84)
<i>Ln(1 + AG)</i>	0.687 (1.74)	0.224 (0.62)	0.945 (2.42)	0.438 (1.21)
stocks	1347	1348	1298	1298
adj- R^2	15.5%	16.7%	13.0%	16.7%

Note: This table reports the results of Fama-MacBeth cross-sectional regressions for realized returns (in excess of the risk-free rate) and ICC (in excess of the yield on 10-year long-term government bonds) on Amihud illiquidity or the *Composite* and various firm characteristics as in Tables 4 and 6 except that we use alternate measures of ICC. The first column uses an ICC based on Easton (2004), and the second column uses an ICC based on Ohlson and Juettner-Nauroth (2005). Variables are defined in Tables 1 and 5. All independent variables are winsorized at 0.5% and 99.5% levels. All variables are lagged by 2 months in return regressions and by 1 month in ICC regressions. *Return11* is lagged by 2 months and *Return1* is lagged by 1 month in all regressions. All coefficients are multiplied by 100. *t* statistics are calculated using the Newey and West (1987) correction with 36 lags for ICC regressions. The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000. The sample period is 1977–2021.

TABLE A4 Fama-MacBeth regressions for realized returns and ICC with additional beta controls.

<i>XILLIQ</i>	<i>Ln(Amihud)</i>		<i>Composite</i>	
	<i>Return</i>	<i>ICC</i>	<i>Return</i>	<i>ICC</i>
<i>Constant</i>	−0.749 (−1.48)	0.614 (0.28)	0.130 (0.34)	−7.543 (−2.92)
<i>XILLIQ</i>	0.118 (1.38)	−0.363 (−2.71)	−0.168 (−4.63)	0.443 (9.21)
<i>Ln(SIZE)</i>	0.043 (0.82)	−0.563 (−5.36)	−0.099 (−3.65)	0.394 (5.07)
<i>Ln(B/M)</i>	0.139 (3.22)	1.244 (4.56)	0.132 (3.05)	1.308 (4.09)
<i>Return11</i>	0.650 (4.28)	−1.581 (−11.19)	0.666 (4.33)	−1.426 (−8.84)
<i>Return1</i>	−3.406 (−8.55)	−3.548 (−21.04)	−3.240 (−8.02)	−3.883 (−16.78)
<i>Beta1</i>	0.092 (0.84)	1.383 (8.40)	0.121 (1.08)	1.583 (9.17)
<i>Beta2</i>	0.031 (0.55)	0.755 (9.20)	0.043 (0.74)	0.816 (6.80)
<i>Beta3</i>	0.120 (1.82)	0.229 (1.69)	0.117 (1.77)	0.307 (1.99)
<i>Beta4</i>	0.011 (0.23)	−0.236 (−4.48)	0.006 (0.12)	−0.223 (−3.25)
<i>Beta5</i>	0.015 (0.34)	0.086 (1.39)	0.018 (0.41)	0.118 (1.61)
<i>GP/AT</i>	0.569 (4.06)	−1.071 (−6.64)	0.549 (3.78)	−1.135 (−5.66)
<i>Ln(1 + AG)</i>	−0.393 (−3.58)	−0.858 (−2.72)	−0.411 (−3.68)	−0.912 (−2.31)
No. of stocks	1345	1042	1295	611
Adj. R^2	8.1%	16.9%	8.3%	17.7%

Note: This table reports the results of Fama-MacBeth cross-sectional regressions for realized returns (in excess of the risk-free rate) and ICC (in excess of the yield on 10-year long-term government bonds) on Amihud illiquidity or the Composite illiquidity and various firm characteristics as in Tables 4 and 6 except that we include five Fama and French (2015) factor betas instead of a single market beta on the right-hand side. Variables are defined in Tables 1 and 5. All independent variables are winsorized at 0.5% and 99.5% levels. All variables are lagged by 2 months in return regressions and by 1 month in ICC regressions. *Return11* is lagged by 2 months and *Return1* is lagged by 1 month in all regressions. All coefficients are multiplied by 100. *t* statistics in parentheses are the standard ones for return regressions and are calculated using the Newey and West (1987) correction with 36 lags for ICC regressions. The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000. The sample period is 1977–2021.

TABLE A5 Fama-MacBeth regressions for ICC: Subsample results.

XILLIQ	1977–1999				2000–2021			
	Ln (Amihud)		Composite		Ln(Amihud)		Composite	
	Return	ICC	Return	ICC	Return	ICC	Return	ICC
Constant	−1.931 (−2.36)	−2.018 (−0.55)	−0.626 (−1.06)	−7.926 (−2.61)	0.669 (0.87)	5.071 (5.50)	0.889 (1.53)	1.819 (2.03)
XILLIQ	0.198 (1.85)	−0.631 (−3.57)	−0.258 (−5.36)	0.964 (10.63)	−0.030 (−0.21)	−0.125 (−1.01)	−0.086 (−1.44)	0.816 (12.68)
Ln(SIZE)	0.102 (1.34)	−0.927 (−5.69)	−0.127 (−3.12)	−0.052 (−0.83)	−0.056 (−0.62)	−0.473 (−5.45)	−0.084 (−2.17)	−0.025 (−0.35)
Ln(B/M)	0.262 (3.92)	1.846 (4.50)	0.250 (3.73)	1.929 (4.84)	0.049 (0.69)	0.716 (5.17)	0.050 (0.70)	0.683 (4.95)
Return11	1.185 (7.43)	−1.416 (−6.49)	1.207 (7.48)	−1.222 (−6.38)	0.129 (0.45)	−1.397 (−5.20)	0.142 (0.50)	−1.445 (−5.51)
Return1	−4.459 (−9.43)	−3.192 (−10.40)	−4.299 (−8.74)	−4.024 (−12.57)	−1.887 (−2.80)	−3.438 (−11.78)	−1.728 (−2.56)	−3.881 (−12.46)
Beta1	0.089 (0.56)	1.564 (5.00)	0.161 (0.98)	1.357 (6.05)	0.027 (0.15)	1.390 (7.67)	0.022 (0.12)	1.275 (7.31)
GP/AT	0.606 (3.71)	−0.745 (−2.74)	0.571 (3.38)	−0.734 (−2.68)	0.547 (2.11)	−1.421 (−8.00)	0.559 (2.05)	−1.392 (−7.62)
Ln (1 + AG)	−0.585 (−4.00)	−1.314 (−2.10)	−0.672 (−4.40)	−1.254 (−1.88)	−0.240 (−1.26)	−0.576 (−2.84)	−0.191 (−1.01)	−0.560 (−2.77)
No. of stocks	1507	1082	1420	1052	1175	1001	1164	995
Adj. R ²	5.9%	15.1%	6.1%	16.2%	7.8%	15.8%	7.9%	17.0%

Note: This table reports the results of Fama-MacBeth cross-sectional regressions for realized returns (in excess of the risk-free rate) and ICC (in excess of the yield on long-term government bonds) on Amihud illiquidity or the Composite illiquidity and various firm characteristics as in Tables 4 and 6 except that we report results for two subsamples of 1977–1999 and 2000–2021. Variables are defined in Tables 1 and 5. All independent variables are winsorized at 0.5% and 99.5% levels. All variables are lagged by 2 months in return regressions and by 1 month in ICC regressions. Return11 is lagged by 2 months and Return1 is lagged by 1 month in all regressions. All coefficients are multiplied by 100. *t* statistics in parentheses are the standard ones for return regressions and are calculated using the Newey and West (1987) correction with 36 lags for ICC regressions. The sample includes NYSE and AMEX stocks with prices between \$5 and \$1000.